D-modules
Lectures by Victor Ginzburg

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Preface

There will be some gaps in explanation, either due to the lecturer's admission or my own lack of understanding. In particular, many "proofs" are sketches of proofs. Gaps due to my own misunderstanding will be indicated by three red question marks: ???. More generally, my own questions about the material will also be in red. Things like "**Question**" will be questions posed by the lecturer. Feel free to reach out to me with explanations.

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D-modules are modules over rings of differential operators. There are three settings:

1. C^∞ case: We can consider differential operators on $\mathbb{R}^n$ of order N, which are expressions
$$u = \sum_{i_1+\cdots+i_n\leq N} a_{i_1,\ldots,i_n}(x)\frac{\partial^N}{\partial x_1^{i_1}\cdots\partial x_n^{i_n}},$$
where the functions $a_\bullet(x)$ are smooth.

2. Complex analytic setting: Similar, but replace $\mathbb{R}$ with $\mathbb{C}$ and require the functions $a_\bullet(x)$ are holomorphic.

3. Algebraic setting: Replace $\mathbb{C}$ with any field and require the functions $a_\bullet(x)$ to be polynomials.

We will work with smooth algebraic varieties over $\mathbb{C}$.

Remark. 1. If X is singular, the ring $\mathcal{D}(X)$ of differential operators might not be noetherian.

2. The theory of differential operators in positive characteristic is very different.

3. We can effectively replace $\mathbb{C}$ by any algebraically closed field of characteristic zero.

Today X is smooth and affine. $\mathcal{O} = \mathcal{O}(X)$ will denote the algebra of regular functions on X. We might not have coordinates on X, so we need to modify our definition of differential operators.

Differential operators of order 0 are just multiplication by some $f \in \mathcal{O}$, denoted m_f. Differential operators of order 1 are $\xi + m_f$, where $f \in \mathcal{O}$ and ξ is a vector field on X, i.e. an element of $T_X = \mathrm{Der}(\mathcal{O}, \mathcal{O})$, i.e. a derivation $\mathcal{O} \to \mathcal{O}$.

Definition 1.1. The algebra $\mathcal{D} = \mathcal{D}(X)$ of differential operators on X is defined as the subalgebra of $\mathrm{End}_{\mathbb{C}}(\mathcal{O})$ generated by the operators of order 0 and 1 defined above.

Note that $\mathcal{O} + T_X$ sits as a subspace in $\mathcal{D}$.

When $X = \mathbb{C}^n$, or more generally is an open subvariety of $\mathbb{C}^n$, then this definition agrees with the previous one using the induced coordinates from $\mathbb{C}^n$.

1.1 Applications of D-modules

Fix $f \in \mathcal{O}$. Let $\mathrm{Ann}(f) = \{u \in \mathcal{D} \mid u(f) = 0\}$. It is a left ideal in $\mathcal{D}$. We define $\mathcal{D}f$ to be $\mathcal{D}/\mathrm{Ann}(f)$, which is a $\mathcal{D}$-module.

We can define $\mathcal{D}f$ for some $f \notin \mathcal{O}$:

1. Let $X = \mathbb{C}$ and $f = e^x$. Certainly f is not regular, but its annihilator is generated by $\frac{d}{dx} - 1$, so we can define $\mathcal{D}e^x = \mathcal{D}/(\frac{d}{dx} - 1)$.

2. Similarly we can define $\mathrm{Ann}(f)$ and hence $\mathcal{D}f$ for $X = \mathbb{C}$ and $f \in C^\infty(\mathbb{R})$.

3. Let $f \in \mathbb{R}[x_1, \dots, x_n]$ and let $U = \{x \in \mathbb{R}^n \mid f(x) \geq 0\}$. Fix $\varphi \in C^\infty(U)$ with compact support. Let $I_\phi(\lambda) = \int_U \varphi(x) f(x)^\lambda dx$ for $\lambda \in \mathbb{C}$. Note that the exponentiation is well-defined because $f(x) \geq 0$. The integral converges when the real part of λ is positive. Then

   Theorem 1.1. *The function $\lambda \mapsto I_\varphi(\lambda)$ extends to a meromorphic function on $\mathbb{C}$ with poles at a finite union of arithmetic progressions of the form $a, a-1, \dots$.*

   Sketch. Say $I_\varphi(\lambda)$ converges for $\lambda > N$ (at the start, we have $N = 0$, but we proceed inductively). We compute

   $$\frac{d}{d\lambda} I_\varphi(\lambda) = \int_U \varphi(x) \log f(x) f(x)^\lambda dx,$$

   which converges, so $I_\varphi(\lambda)$ is holomorphic when $\mathrm{Re}\lambda \geq N$. Our goal is to extend the function to $\mathrm{Re}\lambda \geq N - 1$ inductively. We need:

   Lemma 1.1 (b-function)**.** *There exists $u \in \mathcal{D}(\mathbb{C}^n)[s]$ and $b \in \mathbb{C}[s]$ such that $u(f^{s+1}) = b(s) f^s$.*

   Assuming the lemma, we have

   $$\begin{aligned} I_\varphi(\lambda) = \int_U \varphi(x) f(x)^\lambda dx &= \int_U \varphi(x) \frac{1}{b(\lambda)} u(f(x)^{\lambda+1}) dx \\ &= \frac{1}{b(\lambda)} \int_U (u^*(\varphi))(x) f(x)^{\lambda+1} dx, \end{aligned}$$

   where at the end we have used essentially integration by parts, and $u^* \in \mathcal{D}(\mathbb{C}^n)[s]$ is the formal adjoint of u. For $\mathrm{Re}\lambda > N-1$ we can extend $I_\varphi(\lambda)$ using the above computation.

Example 1.1 (Examples of b-functions)**.**

1. Let $f(x) = \sum x_i^2$ on $\mathbb{C}^n$. Then we can take $u = \sum \frac{\partial^2}{\partial x_i^2}$ and $b(s) = 4(s+1)(s + \frac{n}{2})$.

2. Let $f(x) = \det(x)$ on $\mathbb{C}^{n^2} = M_n(\mathbb{C})$. Then for $u = \det(\frac{\partial}{\partial x_{ij}})$ one can compute $u(f^s)$ using Capelli identities and find that $b(s) = \prod_{1 \leq k \leq n} (s+k)$.

3. If $f(x_1, \dots, x_n) = \prod_{1 \le i < j \le n}$, then the b-function is unknown. Can also generalize to arbitrary root systems.

4. If one takes the square of the above function, then the b-function is known, but is very complicated.

1.2 Nearby cycles

Let $f \in \mathbb{C}[x_1, \dots, x_n]$. Let $\mathrm{Sing}(f) = \{x \in \mathbb{C}^n \mid df(x) = 0\}$. Fix $z_0 \in \mathrm{Sing}(f)$. Take $0 < \delta \ll \varepsilon \ll 1$. We define the *link* of f at z_0 to be

$$L = L(f, z_0, \varepsilon, \delta) = \{z \in \mathbb{C}^n \mid \|z - z_0\| \le \varepsilon, \|f(z) - f(z_0)\| = \delta\}.$$

Topologically, L is independent of δ, ε. there's a nice picture but seems hard to TiKz. Let S^1_δ denote the circle of w such that $\|w - f(z_0)\| = \delta$. The family $H_p(L_w)$ forms a locally constant sheaf on S^1_δ. For fixed w, there is a *monodromy operator* $M_{f,p} : H_p(L_w) \to H_p(L_w)$.

Theorem 1.2. *If $b_f(\lambda) = 0$, then $e^{2\pi i \lambda}$ is an eigenvalue of $M_{f,p}$ for some p. Moreover if $\mathrm{Sing}(f) = \{z_0\}$, then all eigenvalues of all $M_{f,p}$ arise in this way.*

1.3 Microlocal geometry

$\mathcal{D}(X)$ is a noncommutative deformation (aka quantization) of $\mathcal{O}(T^*X)$.

For $f \in \mathcal{O}(X)$, we have $\mathrm{Sing}(f) \subset X$. We also have the $\mathcal{D}$-module $\mathcal{D}f$. To a $\mathcal{D}$-module we can define its singular support, which sits in T^*X. The following square commutes:

$$\begin{tikzcd} SS(\mathcal{D}f) \arrow[r, hook] \arrow[d] & T^*X \arrow[d] \\ \mathrm{Sing}(f) \arrow[r, hook] & X \end{tikzcd}$$

The singular support of $\mathcal{D}f$ can contain more information than just $\mathrm{Sing}(f)$.

1.4 de Rham functor

There is a functor from the (derived?) category of $\mathcal{D}(X)$-modules to the (derived?) category of complexes of sheaves of $\mathbb{C}$-vector spaces on X. It restricts to a functor between the (derived?) category of holonomic modules to the category of perverse sheaves. This further restricts to an equivalence of categories between holonomic modules with regular singularities (or complexes with regular holonomic cohomology?) to the category of perverse sheaves.

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2.1 Poincaré-Birkhoff-Witt (PBW) for differential operators

We continue to assume X is smooth affine, and we write $\mathcal{O}, \mathcal{D}$ for $\mathcal{O}(X), \mathcal{D}(X)$. We let T_X denote the space of vector fields.

For $\xi \in T_X$ and $f, g \in \mathcal{O}$, we have the Liebniz formula $\xi(fg) = f\xi(g) + \xi(f)g$. In particular, in terms of operators, $\xi m_f - m_f \xi = m_{\xi(f)}$. Also, for $\eta \in T_X$, we have $\xi\eta - \eta\xi = [\xi, \eta]$, where $[-,-]$ is the (a priori abstract) Lie bracket of vector fields. It follows that $\mathcal{D}_1 = \mathcal{O} + T_X$ is a Lie algebra. $\mathcal{O}$ is an abelian Lie ideal and T_X is a Lie subalgebra. We have a short exact sequence $0 \to \mathcal{O} \to \mathcal{D}_1 \to T_X \to 0$. The formula $\xi m_f - m_f \xi = m_{\xi(f)}$ also shows that $\mathcal{D}_1$ is an $\mathcal{O}$-bimodule, and that the left and right actions agree on $\mathcal{D}_1/\mathcal{O}$.

Definition 2.1. The **order filtration** on $\mathcal{D}$ is given by $\mathcal{D}_0 = \mathcal{O}$, $\mathcal{D}_1 = \mathcal{O} + T_X$, and $\mathcal{D}_n$ is the image of the composition map $\mathcal{D}_1 \otimes_{\mathcal{O}} \cdots \otimes_{\mathcal{O}} \mathcal{D}_1 \to \mathcal{D}$.

We have

- $\bigcup_n \mathcal{D}_n = \mathcal{D}$.
- $\mathcal{D}_n$ is an $\mathcal{O}$-bimodule.
- $\mathcal{D}_m \mathcal{D}_n \subset \mathcal{D}_{m+n}$.

Let $\mathrm{gr}\mathcal{D} = \bigoplus (\mathcal{D}_n/\mathcal{D}_{n-1})$. Let $f \in \mathcal{O}$ and $\xi, \eta \in T_X$. Then $f\xi \equiv \xi f \mod \mathcal{O}$, so $\mathcal{O} = \mathcal{D}_0$ is a central subalgebra of $\mathrm{gr}\mathcal{D}$. Similarly, $\xi\eta \equiv \eta\xi \mod \mathcal{D}_1$, so $T_X = \mathcal{D}_1/\mathcal{D}_0$ is contained in $Z(\mathrm{gr}\mathcal{D})$. We obtain a surjective graded algebra map $\mu : \mathrm{Sym}_{\mathcal{O}} T_X \to \mathrm{gr}\mathcal{D}$.

Theorem 2.1 (PBW for $\mathcal{D}$)**.** *μ is an isomorphism.*

Proof. It suffices to show that μ is injective.

The cotangent bundle T^*X is a vector bundle, so it admits an (attracting) algebraic $\mathbb{C}^\times$-action given by fiberwise scaling. This gives a grading $\mathcal{O}(T^*X) = \bigoplus_{n \geq 0} \mathcal{O}_n(T^*X)$, where $\mathcal{O}_n(T^*X)$ consists of functions that are fiberwise homogeneous of degree n. Now, there is a canonical isomorphism $\nu : \mathrm{Sym}^n_{\mathcal{O}}(T_X) \xrightarrow{\sim} \mathcal{O}_n(T^*X)$, where a product of vectors $\xi_1 \cdots \xi_n$ is sent to the function taking $(x, \alpha) \in T^*X$ to $\prod \langle (\xi_i)_x, \alpha \rangle$.

Now, there is a so-called principal symbol map that we will define in a moment, $\sigma_n : \mathcal{D}_n \to \mathcal{O}_n(T^*X)$, that sends $\mathcal{D}_{n-1}$ to 0, so it descends to a map $\overline{\sigma}_n : \mathcal{D}_n/\mathcal{D}_{n-1} \to \mathcal{O}_n(T^*X)$. We will check that the composite

$$\mathrm{Sym}^n_{\mathcal{O}} T_X \xrightarrow{\mu} \mathcal{D}_n/\mathcal{D}_{n-1} \xrightarrow{\overline{\sigma}_n} \mathcal{O}_n(T^*X) \xrightarrow{\nu^{-1}} \mathrm{Sym}^n_{\mathcal{O}} T_X$$

is the identity, which implies μ is injective.

Now we construct σ_n. Note that $\mathcal{D}$ acts on $\mathcal{O}[[t]]$ by $u(\sum f_i t^i) = \sum u(f_i)t^i$. For fixed $f \in \mathcal{O}$, we have $e^{tf} = \sum \frac{1}{i!} f^i t^i \in \mathcal{O}[[t]]$.

Claim 2.1. *1. For $u \in \mathcal{D}_n$, we have*

$$u(e^{tf}) = (t^n \sigma_n(u,f) + \textit{ lower degree terms})\, e^{tf},$$

where $\sigma_n(u,f), A \in \mathcal{O}$.

2. If $u \in \mathcal{D}_{n-1}$, then $\sigma_n(u,f) = 0$.

3. We have

$$\sigma_n(\xi_1 \cdots \xi_n, f)(x) = \prod \langle (\xi_i)_x, d_x f\rangle = \nu(\xi_1 \cdots \xi_n)(d_x f).$$

Proof of Claim. First we have

$$\xi e^{tf} = t\xi(f)e^{tf} = t\langle \xi, df\rangle e^{tf}.$$

Then we have

$$\xi_1\xi_2 e^{tf} = t^2 \xi_1(f)\xi_2(f)e^{tf} + t\xi_1\xi_2(f)e^{tf} = \left(t^2\langle \xi_1, df\rangle\langle \xi_2, df\rangle + t\xi_1\xi_2(f)\right) e^{tf}.$$

And from here the pattern is clear.

This claim both gives the construction of σ_n and shows that the previous composition is the identity, so we are done. I want to check this in more detail I think

Remark. PBW and its proof work in the C^∞ or complex analytic settings.

2.2 An application to the PBW theorem for Lie algebras

Let $\mathfrak{g}$ be a Lie algebra, and let $\mathcal{U}\mathfrak{g}$ be its universal enveloping algebra. Lecturer: "since this is not a course on Lie algebras, I'm not supposed to explain what it is." $\mathcal{U}\mathfrak{g}$ has a PBW filtration defined in a very similar way to the order filtration. In particular $\mathcal{U}_0\mathfrak{g} = \mathbb{C}$ and $\mathcal{U}_1\mathfrak{g} = \mathbb{C} + \mathfrak{g}$. Furthermore, the relations of $\mathcal{U}\mathfrak{g}$ imply that we get a surjective graded algebra map $\mu : \mathrm{Sym}_{\mathbb{C}}\mathfrak{g} \to \mathrm{gr}\mathcal{U}\mathfrak{g}$. The PBW theorem (for Lie algebras) asserts that μ is an isomorphism. It is not easy, but if we assume that $\mathfrak{g}$ is the Lie algebra of a Lie group or affine algebraic group $\mathfrak{g}$, then it is easy.

Note. It is in fact true that every $\mathfrak{g}$ comes from a G, but the proof is no easier than PBW, and in some cases one uses PBW in the proof.

We can identify $\mathfrak{g}$ with $(T_G)^G$, i.e. left-invariant vector fields on G, as well as with T_eG. (At least one of these is the definition of Lie$G$, depending on your conventions.) We get a map $\mathfrak{g} \to \mathcal{D}(G)^G$ by sending a vector field to its corresponding differential operator, which gives a map $\tau : \mathcal{U}\mathfrak{g} \to \mathcal{D}(G)^G$.

Theorem 2.2. 1. *The map τ is an algebra isomorphism.*

2. *PBW for $\mathcal{U}\mathfrak{g}$ is true.*

Proof. By construction, $\tau(\mathcal{U}_n\mathfrak{g}) \subset \mathcal{D}_n(G)^G$, so there is an induced graded map $\mathrm{gr}(\tau) : \mathrm{gr}\mathcal{U}\mathfrak{g} \to \mathrm{gr}\mathcal{D}(G)^G$. We then have the following commutative diagram:

$$\begin{tikzcd}
\mathrm{Sym}^n_{\mathbb{C}}\mathfrak{g} \arrow[r, "\sim"] \arrow[d, two heads, "\mu_n"'] & \mathrm{Sym}^n_{\mathcal{O}}((T_G)^G) \arrow[r, "\sim"] & (\mathrm{Sym}^n_{\mathcal{O}} T_G)^G \arrow[d, "\sim \text{by PBW for } \mathcal{D}"] \\
\mathcal{U}_n\mathfrak{g}/\mathcal{U}_{n-1}\mathfrak{g} \arrow[r, "\mathrm{gr}_n(\tau)"'] & \mathcal{D}_n(G)^G/\mathcal{D}_{n-1}(G)^G \arrow[r, hook] & (\mathcal{D}_n(G)/\mathcal{D}_{n-1}(G))^G
\end{tikzcd}$$

The right map on the bottom is injective because $\mathcal{D}_n(G)^G \cap \mathcal{D}_{n-1}(G) = \mathcal{D}_{n-1}(G)^G$. It follows that all arrows are isomorphisms, so we get that $\mathrm{Sym}_{\mathbb{C}}\mathfrak{g} \to \mathrm{gr}\mathcal{U}\mathfrak{g}$ and $\mathrm{gr}\mathcal{U}\mathfrak{g} \to \mathrm{gr}\mathcal{D}(G)^G$ are both isomorphisms. The former isomorphism is PBW, and the latter isomorphism implies τ is an isomorphism, so we are done.

We now discuss another formulation of $\mathcal{D}(X)$. By the lecturer's admission, this may be wrong as it was done on the spot. We have $\mathcal{D}(X) = \mathcal{U}_{\mathbb{C}}(\mathcal{O} + T_X)/(f \otimes \xi = f\xi; 1_{\mathcal{D}} = 1_{\mathcal{O}} = 1_{\mathcal{U}})$. The proof uses PBW to show that the associated graded algebras agree.

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3.1 Sheaves of differential operators

Let X be an irreducible affine. Then $\mathcal{O}(X)$ is a domain. For any nonzero $f \in \mathcal{O}(X)$, we have the basic open $X_f = \{x \in X \mid f(x) \neq 0\}$. It is affine with coordinate ring $\mathcal{O}(X)_f = \mathcal{O}(X)[\frac{1}{f}]$.

Now assume X is not necessarily affine. Facts:

1. A sheaf $\mathcal{E}$ of $\mathcal{O}_X$-modules is quasicoherent if and only if for every open affine $U \subset X$ and nonzero $f \in \mathcal{O}(U)$, the map
$$\Gamma(U, \mathcal{E})_f := \mathcal{O}(U_f) \otimes_{\mathcal{O}(U)} \Gamma(U, \mathcal{E}) \to \Gamma(U_f, \mathcal{E}) \tag{1}$$
is an isomorphism. In this case/equivalently, for all affine opens $V \subset U$, there is an isomorphism
$$\mathcal{O}(V) \otimes_{\mathcal{O}(U)} \Gamma(U, \mathcal{E}) \to \Gamma(V, \mathcal{E}).$$

2. If X is affine, then the map
$$\mathrm{Sym}^n_{\mathcal{O}(X)} \Gamma(X, \mathcal{E}) \to \Gamma(X, \mathrm{Sym}^n_{\mathcal{O}_X} \mathcal{E})$$
is an isomorphism.

Continue assuming X is arbitrary. Suppose to every open affine $U \subset X$ we associate an $\mathcal{O}(U)$-module $\mathcal{E}(U)$, such that there are compatible restriction maps $\mathrm{res} : \mathcal{E}(U) \to \mathcal{E}(U_f)$. For $x \in X$ we define $\mathcal{E}_x = \varinjlim \mathcal{E}(U)$, with the colimit taken over open affines U containing x. This gives a sheaf of $\mathcal{O}_X$ modules that we denote by $\mathcal{E}$. If the map 1 is an isomorphism, then $\mathcal{E}$ is quasicoherent. If X is affine, then there is a map $\mathcal{E}(X) \to \Gamma(X, \mathcal{E})$. If the map 1 is an isomorphism, then so is $\mathcal{E}(X) \to \Gamma(X, \mathcal{E})$.

Let T_X be the tangent sheaf on X. If X is affine, we put $\mathrm{Vect}(X) = \Gamma(X, T_X)$. We define $\mathrm{Sym}^n \mathrm{Vect}(X)$ to be $\mathrm{Sym}^n_{\mathcal{O}(X)} \mathrm{Vect}(X) = \Gamma(X, \mathrm{Sym}^n_{\mathcal{O}_X} T_X)$.

Recall that when X is smooth affine, we have a PBW filtration on $\mathcal{D}(X)$ by $\mathcal{O}(X)$-bimodules, and the PBW theorem asserts there are short exact sequences
$$0 \to \mathcal{D}_{n-1}(X) \to \mathcal{D}_n(X) \xrightarrow{\sigma_n} \mathrm{Sym}^n \mathrm{Vect}(X) \to 0.$$

Proposition 3.1. *Let X be smooth affine and irreducible, and let $f \in \mathcal{O}(X)$ be nonzero.*

1. *For all $u \in \mathcal{D}_n(X)$, there is a unique $u_f \in \mathcal{D}_n(X_f)$ such that the map $u_f : \mathcal{O}(X_f) \to \mathcal{O}(X_f)$ extends that of u.*

2. $\sigma_n(u_f) = \sigma_n(u)|_{X_f}$.

3. *The maps*

$$\mathcal{O}(X_f) \otimes_{\mathcal{O}(X)} \mathcal{D}_n(X) \to \mathcal{D}_n(X_f) \leftarrow \mathcal{D}_n(X) \otimes_{\mathcal{O}(X)} \mathcal{O}(X_f)$$

are isomorphisms.

Proof. We induct on n. For $n = 0$ the statement is clear check!. For $n = 1$ we use the fact that vector fields, i.e. derivations, can be extended to localizations via the quotient rule check in more detail!. For $n > 1$, we use linearity to reduce to the case where u is a product of vector fields. As in the $n = 1$ case, we extend u by extending each vector field. This gives existence, but we need uniqueness, since linear combinations are not unique. Suppose we have two extensions u_f, u'_f. Then $u_f - u'_f$ extends $0 \in \mathcal{D}_n(X)$. From the way we constructed extensions, we see that $\sigma_n(u_f) = \sigma_n(u)|_{X_f} = \sigma_n(u'_f)$. Then $u_f - u'_f \in \ker \sigma_n = \mathcal{D}_{n-1}(X_f)$, so by induction it is 0. I have a problem with this proof, which is when we use symbols. We don't know that any two extensions come from taking linear combinations... This shows (1) and (2).

Note. Here's another proof that the only extension of 0 is 0 by induction on the order, due to Dragos Crisan. Let u be an extension of 0, and let $a \in \mathcal{O}$. Then for any n,

$$0 = u(a) = u(f^n a/f^n) = f^n u(a/f^n) + [u, m_{f^n}](a/f^n).$$

The operator $[u, m_{f^n}]$ has an order strictly smaller than that of u, so by induction $[u, m_{f^n}](a/f^n) = 0$. Then we have $0 = f^n u(a/f^n)$, so $u(a/f^n) = 0$, so $u = 0$ as desired.

For (3), introduce the notation $\mathcal{O} = \mathcal{O}(X)$ and $\mathcal{O}_f = \mathcal{O}(X_f)$. The functor $\mathcal{O}_f \otimes_{\mathcal{O}} -$ is exact, so we have the following commutative diagram with exact rows:

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{O}_f \otimes_{\mathcal{O}} \mathcal{D}_{n-1}(X) \arrow[r] \arrow[d] & \mathcal{O}_f \otimes_{\mathcal{O}} \mathcal{D}_n(X) \arrow[r] \arrow[d] & \mathcal{O}_f \otimes_{\mathcal{O}} \mathrm{Sym}^n \mathrm{Vect}(X) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{D}_{n-1}(X_f) \arrow[r] & \mathcal{D}_n(X_f) \arrow[r] & \mathrm{Sym}^n \mathrm{Vect}(X_f) \arrow[r] & 0
\end{tikzcd}$$

The left vertical arrow is an isomorphism by induction, and the right vertical arrow is an isomorphism by general algebraic geometry. Thus the middle vertical arrow is an isomorphism as desired.

As a consequence, for all smooth (irreducible) X, we have a quasicoherent sheaf of left/right $\mathcal{O}_X$ modules $\mathcal{D}_{X,n}$. Furthermore, we have a short exact sequence

$$0 \to \mathcal{D}_{X,n-1} \to \mathcal{D}_{X,n} \to \mathrm{Sym}^n T_X \to 0.$$

By induction on n, using the fact that $\mathcal{D}_{X,0} = \mathcal{O}_X$ and $\mathrm{Sym}^n T_X$ are coherent, we find that $\mathcal{D}_{X,n}$ is in fact coherent. We can then take $\mathcal{D}_X = \bigcup_n \mathcal{D}_{X,n}$, a quasicoherent sheaf of associative algebras. PBW implies $\mathrm{gr}\mathcal{D}_X \cong \mathrm{Sym} T_X$.

3.2 Twisted differential operators (TDO)

Let $\mathcal{L}$ be a line bundle on X. Let $\mathcal{D}_{X,\mathcal{L}} = \mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{D}_X \otimes_{\mathcal{O}_X} \mathcal{L}^{-1}$. Here $\mathcal{L}^{-1}$ is the dual bundle to $\mathcal{L}$, sometimes denoted $\mathcal{L}^\vee$ or $\mathcal{L}^*$.

Claim 3.1. *$\mathcal{D}_{X,L}$ has the structure of a sheaf of associative algebras.*

Proof. Let u_1, u_2 be local sections of $\mathcal{D}_X$. Let α_1, α_2 be local sections of $\mathcal{L}$. Let β_1, β_2 be local sections of $\mathcal{L}^{-1}$. We define $(\alpha_1 u_1 \beta_1)(\alpha_2 u_2 \beta_2)$ to be $\alpha_1(u_1 m_{\langle \beta_1, \alpha_2 \rangle}) u_2)\beta_2$.

There is also a natural action of $\mathcal{D}_{X,\mathcal{L}}$ on $\mathcal{L}$, given on local sections by $(\alpha_1 u \beta)\alpha_2 = u(\langle \beta, \alpha_2 \rangle)\alpha_1$.

Replacing $\mathcal{D}_X$ by $\mathcal{D}_{X,n}$ in the definition of $\mathcal{D}_{X,\mathcal{L}}$ gives a sheaf we call $\mathcal{D}_{X,\mathcal{L},n}$, giving a PBW filtration on $\mathcal{D}_{X,\mathcal{L}}$. We have

$$\mathrm{gr}\mathcal{D}_{X,\mathcal{L}} = \mathcal{L} \otimes_{\mathcal{O}_X} \mathrm{gr}\mathcal{D}_X \otimes_{\mathcal{O}_X} \mathcal{L}^{-1} = \mathrm{gr}\mathcal{D}_X,$$

where we can cancel $\mathcal{L}$ and $\mathcal{L}^{-1}$ since $\mathrm{gr}\mathcal{D}_X$ is an ordinary sheaf of $\mathcal{O}_X$ modules, i.e. there is no distinction between left and right.

Definition 3.1. A **$\mathcal{D}_X$-module** is a quasicoherent sheaf $\mathcal{M}$ such that for all open $U \subset X$, we have an action

$$\Gamma(U, \mathcal{D}_X) \times \Gamma(U, \mathcal{M}) \to \Gamma(U, \mathcal{M}).$$

Recall $\mathcal{D}_X = \mathcal{U}(\mathcal{O}_X + T_X)/(f \otimes u = fu \forall f \in \mathcal{O}_X, u \in \mathcal{D}_{X,1}; 1_{\mathcal{U}} = 1_{\mathcal{O}_X})$. Thus, a $\mathcal{D}_X$-module is the same as a quasicoherent sheaf $\mathcal{M}$ with a "flat connection", i.e. maps ∇_ξ for $\xi \in T_X$ acting on local sections of $\mathcal{M}$ such that $\nabla_\xi(fm) = f\nabla_\xi m + \xi(f)m$ and $\nabla_{[\xi,\eta]} = [\nabla_\xi, \nabla_\eta]$.

For a $\mathcal{D}_X$-module $\mathcal{M}$, $\Gamma(X, \mathcal{M})$ is a module over $\Gamma(X, \mathcal{D}_X) = \mathcal{D}(X)$. In particular, $\Gamma(X, -)$ gives a functor from $\mathcal{D}_X$-modules to $\mathcal{D}(X)$-modules. It has a left adjoint called the localization functor Loc, given by $\mathrm{Loc}(M) = \mathcal{D}_X \otimes_{\mathcal{D}(X)} M$. In particular, for an affine open $U \subset X$, we have $\Gamma(U, \mathrm{Loc}(M)) = \mathcal{D}(U) \otimes_{\mathcal{D}(X)} M$.

If X is affine, $\Gamma(X, -)$ is an equivalence $\mathrm{QCoh}(X) \to \mathcal{O}(X)$-mod with inverse $M \mapsto \mathcal{O}_X \otimes_{\mathcal{O}(X)} M$. Using the map 1, one can show that this restricted map from $\mathcal{D}_X$-modules to $\mathcal{D}(X)$-modules is also an equivalence (when X is affine).

3.3 Beilinson-Bernstein localization theorem, the statement

Let G be a connected semisimple algebraic group with fixed Borel subgroup B. Let $\mathcal{B} = G/B$ be the flag variety.

Theorem 3.1 (Beilinson-Bernstein Localization Theorem)**.** *$\Gamma(\mathcal{B}, -) : \mathcal{D}_{\mathcal{B}}$-mod $\to \mathcal{D}(\mathcal{B})$-mod is an equivalence with inverse* Loc.

Remark. A similar statement holds for all partial flag varieties G/P with P parabolic. In particular, it holds for projective spaces. Historically, this statement was first shown for projective spaces in a much simpler way.

4 Apr 2

4.1 Beilinson-Bernstein localization theorem, the proof (part 1)

We start the proof of Theorem 3.1.

4.1.1 Setup

Let G be a connected semisimple algebraic group with fixed Borel subgroup B_0. Let $\mathcal{B} = G/B_0$ be the flag variety. Let D denote the category of $\mathcal{D}_{\mathcal{B}}$-modules. We will often abbreviate $\Gamma(\mathcal{B}, -)$ to just $\Gamma(-)$.

4.1.2 Today's goal

There are two key statements:

1. If $0 \neq \mathcal{M} \in \mathsf{D}$, then $\Gamma(\mathcal{M}) \neq 0$.
2. For all $\mathcal{M} \in \mathsf{D}$ and $i > 0$, we have $H^i(\mathcal{M}) = 0$.

These statements imply that Γ is exact on D. We will then use this to show that any $\mathcal{M} \in \mathsf{D}$ is generated by global sections.

4.1.3 Step 1

Fix a maximal torus $T \subset B_0$. We use $\lambda \in X^*(T) = \mathrm{Hom}(T, \mathbb{C}^\times)$ to denote a character of T.

Fix V, a finite dimensional irreducible representation of G. There is a unique 1-dimensional subspace $\ell(B_0) \subset V$ called the highest weight space such that B_0 acts on $\ell(B_0)$ via a dominant character λ.

Note that G/B_0 can be identified with the set of Borels in G, via $gB_0 \mapsto B = gB_0g^{-1}$. Set $\ell(B) = g\ell(B_0)$. The collection of lines $\ell(B)$ give a line bundle on $\mathcal{B}$. We let $\mathcal{O}(\lambda)$ be the corresponding coherent sheaf of local sections. In particular, there is a map $\iota : \mathcal{B} \to \mathbb{P}(V)$ given by $B \mapsto \ell(B)$, and $\mathcal{O}(\lambda) = \iota^*\mathcal{O}(-1)$. We say λ is **regular** if ι is an embedding; in this case, $\mathcal{O}(-\lambda) = \iota^*\mathcal{O}(1)$ is very ample.

4.1.4 Step 2

By Lie's theorem, we may choose a B_0-stable flag in V starting with $\ell(B_0)$, i.e. subspaces $\ell(B_0) = F_1 \subset F_2 \subset \cdots \subset F_n = V$ where each F_i is B_0-stable and $\dim F_i = i$. Then B_0 acts on F_i/F_{i-1} by a character $\lambda_i \in X^*(T)$. The characters λ_i are the weights of V. We let λ^* denote λ_n; it is the lowest weight of V.

For $B = gB_0g^{-1}$ another Borel, we define $F_i(B) = gF_i$. Then $F_\bullet(B)$ is a B-stable flag in V. For a fixed i, the i-dimensional spaces $F_i(B)$ for $B \in \mathcal{B}$ give a rank i vector bundle on $\mathcal{B}$, which we denote by $F_i(\mathcal{B})$. We obtain a flag of subvector bundles of $F_n(\mathcal{B}) = V \times \mathcal{B}$. We let $\mathcal{F}_i(\mathcal{B})$ denote the corresponding locally free sheaf to each $F_i(\mathcal{B})$. In particular, $\mathcal{F}_1(\mathcal{B}) = \mathcal{O}(\lambda)$ and $\mathcal{F}_n(\mathcal{B}) = V \otimes_{\mathbb{C}} \mathcal{O}_{\mathcal{B}}$. We note that $F_i(\mathcal{B})/F_{i-1}(\mathcal{B})$ is a line bundle whose corresponding sheaf is $\mathcal{O}(\lambda_i)$. Note that we did not define $\mathcal{O}(\nu)$ for an arbitrary character ν, but we will do this next time.

4.1.5 Step 3

Let $\mathcal{M} \in \mathsf{D}$. For any $\nu \in X^*(T)$, let $\mathcal{M}(\nu) = \mathcal{O}(\nu) \otimes_{\mathcal{O}_{\mathcal{B}}} \mathcal{M}$. This is a quasicoherent $\mathcal{O}_{\mathcal{B}}$-module. We also define $\mathcal{F}_i(\mu) = \mathcal{F}_i(\mathcal{B}) \otimes_{\mathcal{O}_{\mathcal{B}}} \mathcal{M}$. We get a flag $\mathcal{M}(\lambda) = \mathcal{F}_1(\mathcal{M}) \subset \cdots \subset \mathcal{F}_n(\mathcal{M}) = V \otimes_{\mathbb{C}} \mathcal{M}$. Since the $\mathcal{F}_i(\mathcal{B})$ are locally free, tensoring is exact, so $\mathcal{F}_i(\mathcal{M})/\mathcal{F}_{i-1}(\mathcal{M}) = \mathcal{M}(\lambda_i)$.

We now have two important short exact sequences. The first is

$$0 \to \mathcal{F}_{n-1}(\mathcal{M}) \to V \otimes_{\mathbb{C}} \mathcal{M} \to \mathcal{M}(\lambda^*) \to 0. \tag{2}$$

For the second, we first note that $V \otimes_{\mathbb{C}} \mathcal{M}(-\lambda)$ has a filtration by $\mathcal{F}_i(\mathcal{M})(-\lambda)$, and $\mathcal{F}_1(\mathcal{M})(-\lambda) = \mathcal{M}$. The second short exact sequence is

$$0 \to \mathcal{M} \to V \otimes_{\mathbb{C}} \mu(-\lambda) \to \mathcal{F}_n(\mathcal{M})(-\lambda)/\mathcal{M} \to 0. \tag{3}$$

4.1.6 Key lemma

Note that, so far, we have not used that $\mathcal{M}$ is a D-module. The following lemma is where this is used.

Lemma 4.1 (Key Lemma)**.** *If $\mathcal{M} \in \mathsf{D}$, then the short exact sequences 2 and 3 split as sheaves of $\mathbb{C}$-vector spaces on $\mathcal{B}$.*

We will prove the lemma next time.

4.1.7 Step 4

We now show that higher cohomology of $\mathcal{M} \in \mathsf{D}$ vanishes.

Fix $i > 0$. Let $b : \mathcal{N} \hookrightarrow \mathcal{M}$ be an embedding of a coherent subsheaf. All the filtration business from before can be done with $\mathcal{N}$, since we never used the D-module structure. We have a commuting square:

$$\begin{tikzcd}
\mathcal{N} \arrow[r, "a"] \arrow[d, hook, "b"'] & V \otimes_{\mathbb{C}} \mathcal{N}(-\lambda) \arrow[d, "\widetilde{b}"] \\
\mathcal{M} \arrow[r, "\widetilde{a}"'] & V \otimes_{\mathbb{C}} \mathcal{M}(-\lambda)
\end{tikzcd}$$

This induces a commuting square on cohomology:

$$\begin{tikzcd}
H^i(\mathcal{N}) \arrow[r, "H^i(a)"] \arrow[d, hook, "H^i(b)"'] & H^i(V \otimes_{\mathbb{C}} \mathcal{N}(-\lambda)) \arrow[r, equal] \arrow[d, "H^i(\widetilde{b})"] & V \otimes_{\mathbb{C}} H^i(N(-\lambda)) \\
H^i(\mathcal{M}) \arrow[r, "H^i(\widetilde{a})"'] & H^i(V \otimes_{\mathbb{C}} \mathcal{M}(-\lambda)) \arrow[r, equal] & V \otimes_{\mathbb{C}} H^i(M(-\lambda))
\end{tikzcd}$$

Now we choose λ large enough so that $\mathcal{O}(-\lambda)$ is very ample and $H^i(\mathcal{N}(-\lambda)) = 0$. Thus $H^i(\widetilde{a}) \circ H^i(b) = 0$. By Lemma 4.1, $\widetilde{a}$ is an embedding of a direct summand, so that $H^i(\widetilde{a})$ is injective. Thus $H^i(b) = 0$.

We have shown that for any coherent subsheaf $\mathcal{N} \subset \mathcal{M}$, the induced map $H^i(\mathcal{N}) \to H^i(\mathcal{M})$ is 0. But $\mathcal{M}$ is a direct limit of its coherent subsheaves, and H^i commutes with direct limits, so we find $H^i(\mathcal{M}) = 0$.

4.1.8 Step 5

We now show that the global sections of a nonzero $\mathcal{M} \in \mathsf{D}$ is nonzero.

There is a nonzero coherent subsheaf $\mathcal{N} \subset \mathcal{M}$. Choose λ such that $\mathcal{O}(\lambda^*)$ is very ample and $\Gamma(\mathcal{N}(\lambda^*)) \neq 0$. Then the injection $\Gamma(\mathcal{N}(\lambda^*)) \hookrightarrow \Gamma(\mathcal{M}(\lambda^*))$ shows that $\Gamma(\mathcal{M}(\lambda^*))$. By Lemma 4.1, we also have an injection $\Gamma(\mathcal{M}(\lambda^*)) \hookrightarrow \Gamma(V \otimes_{\mathbb{C}} \mathcal{M}) = V \otimes_{\mathbb{C}} \Gamma(\mathcal{M})$. Thus $\Gamma(\mathcal{M}) \neq 0$ as desired.

4.1.9 Step 6

We now show that any $\mathcal{M} \in \mathsf{D}$ is generated by global sections.

Let $M = \Gamma(\mathcal{M})$; it is a $\mathcal{D}(\mathcal{B})$-module. Recall that Γ has a left adjoint $\mathrm{Loc} : N \mapsto \mathcal{D}_{\mathcal{B}} \otimes_{\mathcal{D}(\mathcal{B})} N$. The adjunction gives a map $\alpha : \mathrm{Loc}(M) \to \mathcal{M}$. By definition, $\mathcal{M}$ is generated by global sections means that α is surjective. Let $\mathcal{M}'$ be the image of α. The identity $M \to M$ factors as

$$M \xrightarrow{m \mapsto 1 \otimes m} \Gamma(\mathrm{Loc}(M)) \xrightarrow{\Gamma(\alpha)} \Gamma(\mathcal{M}') \hookrightarrow \Gamma(\mathcal{M}) = M.$$

Thus implies that $\Gamma(\mathcal{M}') \xrightarrow{\sim} \Gamma(\mathcal{M})$. Now let $\mathcal{M}''$ be the cokernel of α, so we have a short exact sequence

$$0 \to \mathcal{M}' \to \mathcal{M} \to \mathcal{M}'' \to 0.$$

From steps 4 and 5, we know $\Gamma(-)$ is exact, so we get an exact sequence

$$0 \to \Gamma(\mathcal{M}') \to \Gamma(\mathcal{M}) \to \Gamma(\mathcal{M}'') \to 0.$$

But the map $\Gamma(\mathcal{M}') \to \Gamma(\mathcal{M})$ is an isomorphism, so $\Gamma(\mathcal{M}'') = 0$. By step 5, this means that $\mathcal{M}'' = 0$, as desired.

4.2 Some setup for next time

Let an algebraic group G act on a variety X. Let $E \xrightarrow{p} X$ be a vector bundle. We say the bundle is G-equivariant if

1. G acts on E so that p is an equivariant map.
2. The induced maps $E_x \to E_{gx}$ on fibers are linear.

Let $\mathcal{E}$ be the sheaf of sections of E, and let $\mathfrak{g} = \mathrm{Lie} G$. There is a Lie derivative map which associates to $\xi \in \mathfrak{g}$ a map $\nabla_\xi : \mathcal{E} \to \mathcal{E}$. In the analytic setting, $\nabla_\xi(s) = \frac{d}{dt}(e^{-t\xi}(s))|_{t=0}$. There is an algebraic way to make sense of this as well.

5 Apr 7

Notes for this day provided by Léo Gratien.

5.1 Example of SL_2

- $G = SL_2$ acts on $\mathbb{C}^2$ and
$$N = \left\{ \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \right\} = \mathrm{Stab}_G \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$
- $G/N = \mathbb{C}^2 \setminus 0$ is quasi-affine. By Hartogs $\mathcal{O}(G/N) = \mathbb{C}[x, y]$.
- $G \times T$ acts on G/N by left and right actions.
- $T = \mathbb{C}^* \subset G/N$ acts via $t : v \mapsto t^{-1}v$.
- $\mathfrak{t}$ is 1-dimensional and h maps to the Euler vector field, $-Eu = x\partial_x + y\partial_y$.
- $\mathfrak{sl}_2$ acts via
$$e \mapsto x\partial_y, \qquad f \mapsto y\partial_x, \qquad h \mapsto x\partial_x - y\partial_y.$$
This gives
$$\mathfrak{g} \times \mathfrak{t} \longrightarrow \mathrm{Vect}(G/N).$$
and in turn an algebra map
$$U(\mathfrak{g} \times \mathfrak{t}) = U(\mathfrak{g}) \otimes U(\mathfrak{t}) \longrightarrow \mathcal{D}(G/N).$$

5.2

Let $\mathcal{D}(G/N)^G$ be the subalgebra of right-invariant translations for G.

Lemma 5.1.
$$\alpha : U(\mathfrak{t}) \xrightarrow{\sim} \mathcal{D}(G/N)^G$$
is an isomorphism.

Proof. Look at the associated graded:
$$\mathrm{gr}\big(U(\mathfrak{t})\big) = \mathbb{C}[\mathfrak{t}^*] \longrightarrow \mathrm{gr}\big(\mathcal{D}(G/N)^G\big) \hookrightarrow \mathrm{gr}\big(\mathcal{D}(G/N)\big)^G$$
with
$$\mathrm{gr}\big(\mathcal{D}(G/N)\big)^G \cong \mathcal{O}\big(T^*(G/N)\big)^G.$$
We will identify the geometric incarnation of the latter with the moment map
$$T^*(G/N)//G \longrightarrow \mathfrak{t}^*.$$
Recall that $T^*(G/N) = G \times^N T_e^*$; hence $\mathcal{O}\big(T^*(G/N)\big)^G = \mathcal{O}(T_e^*)^N$. Moreover we have a map
$$\phi : T_e^* = (\mathfrak{g}/\mathfrak{n})^* \xrightarrow{\mathrm{res}} (\mathfrak{b}/\mathfrak{n})^* = \mathfrak{t}^*.$$
Observe :

1. The above construction coincides with the map $\mathrm{gr}(\alpha)$:

$$T^*(G/N) = G \times^N T_e^* \longrightarrow \mathfrak{t}^*, \qquad (gN/N, \alpha) \longmapsto \alpha|_{\mathfrak{b}/\mathfrak{n}},$$

2. An algebra isomorphism

$$\phi^* : \mathbb{C}[\mathfrak{t}^*] \longrightarrow \mathbb{C}[(\mathfrak{g}/\mathfrak{n})^*]^N.$$

Justification of 2. Using a Killing form, one identifies

$$\mathfrak{t}^* \cong \mathfrak{t}, \qquad (\mathfrak{g}/\mathfrak{n})^* \cong \mathfrak{n}^\perp \cong \mathfrak{b},$$

and then $\phi : \mathfrak{b} \to \mathfrak{t}$ is given, for sufficiently general $h \in \mathfrak{t}$, by

$$N \cdot h \longmapsto h.$$

5.3 Harish-Chandra homomorphism

Let $Z(\mathfrak{g}) \subset U(\mathfrak{g})$ be the center. The *Harish-Chandra homomorphism* is the algebra map

$$Z(\mathfrak{g}) \xrightarrow{hc} \mathcal{D}(G/N)^G \cong U(\mathfrak{t}).$$

Remark. The right action of the center is given by the left action.

5.4

$$p : G/N \xrightarrow{T} \mathcal{B}$$

is a principal T-bundle. We have an action of T on $G/N \times \mathbb{C}_\alpha$, for $\alpha \in X^*(T)$:

$$t \cdot (gN/N, z) = \big(gt^{-1}N/N, \alpha(t)z\big).$$

It gives rise to a line bundle $L_\alpha := G/N \times^T \mathbb{C}_\alpha$ on $\mathcal{B}$.

5.5

G acts on L_α by

$$g_0 : (gN/N, z) \longmapsto (g_0 gN/N, z).$$

Local sections of L_α give a sheaf $\mathcal{O}(\alpha)$.

Remark. One see the lattice of character as linear functionals on the Lie torus

$$X^*(T) \hookrightarrow \mathfrak{t}^*, \qquad \xi \in \mathfrak{t} \longmapsto \xi_{\mathcal{B}} \in \mathrm{Vect}(G/N).$$

5.6 Description of the locally free sheaf $\mathcal{O}(\alpha)$.

It is a $\mathcal{O}_{\mathcal{B}}$-module of rank 1. Let $U \subset \mathcal{B}$ be an open affine. Then

$$\Gamma(U, \mathcal{O}(\alpha)) = \left\{ f : p^{-1}(U) \to \mathbb{C} \;\middle|\; f(gt\mathcal{B}/\mathcal{B}) = \alpha(t)\, f(g\mathcal{B}/\mathcal{B}) \right\}.$$

Recall the TDO algebra $\mathcal{D}_\lambda = \mathcal{O}(\lambda) \otimes \mathcal{D}_{\mathcal{B}} \otimes \mathcal{O}(-\lambda)$.
The projection p entails

$$\begin{tikzcd}
\mathcal{D}\big(p^{-1}(U)\big)^T \arrow[rr] & & \mathcal{D}_\lambda(U) \\
U(\mathfrak{t}) \arrow[u, hook] \arrow[r] & Z\Big(\mathcal{D}\big(p^{-1}(U)\big)^T\Big) \arrow[ul, hook] &
\end{tikzcd}$$

with the central map coming from $\xi \in \mathfrak{t} \longmapsto \lambda(\xi) \cdot \mathrm{Id}$.

Proposition 5.1.

$$\Gamma(U, \mathcal{D}_\lambda) = \mathcal{D}\big(p^{-1}(U)\big)^T / I,$$

where I is the two-sided ideal generated by $\xi - \lambda(\xi), \qquad \xi \in \mathfrak{t}$.

Now $z \in Z(\mathfrak{g})$ acts on $\mathcal{D}_\lambda(U)$. How? Through the HC homomorphism:

$$hc(z) \in U(\mathfrak{t}) = \mathbb{C}[\mathfrak{t}^*], \qquad hc(z)(\lambda) \in \mathbb{C}.$$

5.7 A concrete statement

Denote $\chi_\lambda : Z(\mathfrak{g}) \to \mathbb{C}, z \mapsto hc(z)(\lambda)$, the algebra character.
$\ker(\chi_\lambda)$ is a maximal ideal in $Z(\mathfrak{g})$ and gives

$$I_\lambda := U\mathfrak{g} \cdot \ker \chi_\lambda = \ker \chi_\lambda \cdot U\mathfrak{g},$$

a two-sided ideal.

Theorem 5.1 (Beilinson-Bernstein, part 1)**.** *1. The map*

$$U\mathfrak{g} \longrightarrow \Gamma(\mathcal{B}, \mathcal{D}_\lambda)$$

factors through an algebra isomorphism

$$U\mathfrak{g}/I_\lambda \xrightarrow{\sim} \Gamma(\mathcal{B}, \mathcal{D}_\lambda).$$

2. In particular,

$$\mathcal{D}(G/B) \cong U\mathfrak{g}/\ker \chi_0\, U\mathfrak{g} \cong U\mathfrak{g}/(U\mathfrak{g}) \cdot \mathfrak{g}$$

(augmentation ideal).

5.7.1

We now prove this theorem. Taking associated graded, we look at

$$\operatorname{gr}(U\mathfrak{g}/I_\lambda) \xrightarrow{?} \operatorname{gr}\big(\mathcal{D}_\lambda(G/B)\big).$$

But one has

$$\begin{tikzcd}
\operatorname{gr}(U\mathfrak{g})/\operatorname{gr}(U\mathfrak{g})\operatorname{gr}(\ker\chi_\lambda) \arrow[r] \arrow[d, "\phi"'] & \operatorname{gr}(U\mathfrak{g}/I_\lambda) \arrow[d] \\
\Gamma(G/B, \operatorname{gr}\mathcal{D}) \cong \Gamma(G/B, \operatorname{gr}\mathcal{D}_\lambda) & \operatorname{gr}\big(\mathcal{D}_\lambda(G/B)\big) \arrow[l]
\end{tikzcd}$$

We will show that the natural map ϕ is an isomorphism.

Since $\ker\chi_\lambda$ is generated by $\{z - \chi_\lambda(z), z \in Z\mathfrak{g}\}$, its graded $\operatorname{gr}(\ker\chi_\lambda)$ inside $\operatorname{Sym}(\mathfrak{g}) = \mathbb{C}[\mathfrak{g}^*]$ is given by

$$J = \{p \in \mathbb{C}[\mathfrak{g}^*]^G \mid p(0) = 0\}.$$

Therefore

$$\operatorname{gr}(U\mathfrak{g})/\operatorname{gr}(U\mathfrak{g})\operatorname{gr}(\ker\chi_\lambda) \cong \mathbb{C}[\mathfrak{g}^*]/\mathbb{C}[\mathfrak{g}^*]\cdot J = \mathcal{O}(N).$$

Next time we will give more details on this proof.

6 Apr 9

Notes for this day provided by Léo Gratien.

6.1 Reminders on central characters

Assume G is a connected Lie algebraic group over $\mathbb{C}$.

Proposition 6.1. *1. If G is connected,*

$$Z(\mathfrak{g}) = [U\mathfrak{g}]^G.$$

2. With respect to the PBW grading,

$$\operatorname{gr}(U\mathfrak{g}) = \operatorname{Sym}\mathfrak{g} = \mathbb{C}[\mathfrak{g}^*], \qquad \operatorname{gr}(Z\mathfrak{g}) = [\operatorname{Sym}\mathfrak{g}]^G = \mathbb{C}[\mathfrak{g}^*]^G.$$

3. $U\mathfrak{g}$ has countable dimension over $\mathbb{C}$. Schur's lemma gives

$$\chi : Z\mathfrak{g} \longrightarrow \mathbb{C},$$

acting by scalars $z \longmapsto \chi(z) \cdot \operatorname{Id}$.

4. $\ker\chi$ is a codimension 1 ideal of $Z\mathfrak{g}$. The action factors through

$$U_\chi = U\mathfrak{g}/U\mathfrak{g} \cdot \ker\chi.$$

Remark. 1. We have

$$\operatorname{gr}(U\mathfrak{g}) \supset \operatorname{gr}(Z\mathfrak{g}) \supset \operatorname{gr}(\ker\chi),$$

that is,

$$\mathbb{C}[\mathfrak{g}^*] \supset \mathbb{C}[\mathfrak{g}^*]^G \supset \mathbb{C}[\mathfrak{g}^*]^G_+.$$

It follows from PBW, cf. last time.

2. The graded $\operatorname{gr}(\ker\chi)$ is independent of χ.

Geometrically if $\varpi : \mathfrak{g}^* \longrightarrow \mathfrak{g}^*//G$, then $\operatorname{Spec}\big(\mathbb{C}[\mathfrak{g}^*]^G_+\big)$ is its fiber at 0.

6.1.1

The map

$$\mathcal{O}\big(\varpi^{-1}(0)\big) = \mathcal{O}(N) \longrightarrow \operatorname{gr}(U_\chi)$$

can (when it is an isomorphism!) be seen as a deformation of $\mathcal{O}\big(\varpi^{-1}(0)\big)$.

6.1.2

But in general ϖ does not behave well:

- $\varpi^{-1}(0)$ may be bad,
- $\mathcal{O}\big(\varpi^{-1}(0)\big) \to \operatorname{gr}(U_\chi)$ need not be an isomorphism.

6.2 The reductive case

From now on, G is connected reductive. Here are two main results of Kostant.

Theorem 6.1 (Kostant). *1. $\mathbb{C}[\mathfrak{g}^*]$ is a free module of infinite rank over $\mathbb{C}[\mathfrak{g}^*]^G$.*

2. $U\mathfrak{g}$ is a free $Z\mathfrak{g}$-module.

3. $\varpi^{-1}(0) = N$, the nilpotent cone.

4. U_χ is a flat deformation of $\varpi^{-1}(0) = N$, i.e.

$$\mathcal{O}(N) \xrightarrow{\sim} \mathrm{gr}(U_\chi)$$

is an isomorphism for all χ.

6.2.1

Last time we constructed the HC homomorphism

$$hc : Z\mathfrak{g} \longrightarrow \mathcal{D}(G/N)^G = U\mathfrak{t} \cong \mathbb{C}[\mathfrak{t}^*].$$

Theorem 6.2 (Harish-Chandra). *1. hc is injective.*

2.

$$\mathrm{Im}(hc) = \mathbb{C}[\mathfrak{t}^*]^W$$

for the dot action

$$W : \mathfrak{t} \longrightarrow \mathfrak{t}, \qquad \lambda \longmapsto w(\lambda - \rho) + \rho.$$

[ρ is the only fixed point of the dot action.]

3. One gets a description of central characters in terms of W-orbits on $\mathfrak{t}^*$, or equivalently semisimple G-orbits in $\mathfrak{g}^*$.

6.3 Proof of BB, part 1

Passing to graded, it suffices to prove an isomorphism

$$\mathrm{gr}(U_{\chi_\lambda}) \longrightarrow \mathrm{gr}\big(\mathcal{D}_\lambda(G/B)\big).$$

By Kostant,

$$\mathcal{O}(N) \longrightarrow \mathcal{O}\big(T^*(G/B)\big)$$

should be identified with the graded map. To prove that the graded map is an isomorphism, we identify

$$\mu^* = \mathrm{gr}(\lambda) : T^*(G/B) \to N.$$

We assume the following lemma.

Lemma 6.1. *1. μ is the moment map of the Springer resolution.*

2. $\mathrm{Im}(\mu) = N$ *and μ is a birational isomorphism, proper.*

3. (Kostant) N is normal.

Now one observes :

- μ^* is an embedding of domains. (by 1), 3))
- $\mathrm{Frac}(\mu^*)$ is an isomorphism. (by 2))
- $\mu_* \mathcal{O}_{T^*(G/B)}$ is a coherent sheaf on N. (by 1))

Then $\Gamma\big(N, \mu_* \mathcal{O}_{T^*(G/B)}\big)$ is finite over $\mathcal{O}(N)$, and also $\Gamma\big(N, \mu_* \mathcal{O}_{T^*(G/B)}\big) = \Gamma\big(T^*(G/B), \mathcal{O}_{T^*(G/B)}\big)$.
Hence $\mathcal{O}\big(T^*(G/B)\big)$ is an integral extension of $\mathcal{O}(N)$. By (3), we deduce that μ^* is an isomorphism. This ends the proof of BB, part 1.

6.4 Beilinson-Bernstein, part 2

Before stating the second theorem of BB, some preparations.

6.5

Let $\mu \in X^*(T)$ be an anti-dominant weight. Recall

$$\mathcal{D}_\mu\text{-mod} \underset{\mathrm{Loc}(-)}{\overset{\Gamma(\mathcal{B},-)}{\rightleftarrows}} \mathcal{D}_\mu(G/B)\text{-mod}$$

where

$$\mathrm{Loc}(-) = \mathcal{D}_\mu \otimes_{\mathcal{D}_\mu(G/B)} (-).$$

Remark. $\mu = 0$ is anti-dominant.

Theorem 6.3 (Beilinson-Bernstein, part 2)**.** *The functors above define an equivalence of categories*

$$U_{\chi_\mu}\textit{-mod} \longrightarrow \mathcal{D}_\mu\textit{-mod}.$$

6.5.1

We turn to the proof. The key step is the following.

Let V be a finite-dimensional irreducible representation of $\mathfrak{g}$, of highest weight λ, lowest weight λ^*, i.e. dominant (resp. anti-dominant). We constructed:

$$\begin{aligned} &(\mathrm{I}) \qquad V \otimes \mathcal{O}_{G/B} \twoheadrightarrow \mathcal{O}(\lambda^*), \\ &(\mathrm{II}) \qquad \mathcal{O}_{G/B} = \mathcal{O} \hookrightarrow V \otimes \mathcal{O}(-\lambda). \end{aligned}$$

For every $\mathcal{O}$-module M, applying these gives

$$\begin{aligned} &V \otimes M \longrightarrow \mathcal{O}(\lambda^*) \otimes M =: M(\lambda^*), \\ &M \longrightarrow V \otimes M(-\lambda). \end{aligned}$$

Lemma 6.2 (Lemme clef)**.** *Let $\mu \in X^*(T)$ be anti-dominant. If M is a $\mathcal{D}_\lambda$-module, then (I) and (II) split as $\mathbb{C}$-vector spaces (without any extra structure).*

7 Apr 14

7.1 Splitting lemma

We only prove one of the two splittings, the second is similar.

Consider the case $\mu = 0$ so $\mathcal{D}_\mu = \mathcal{D}_\mathcal{B}$. Fix a finite dimensional irreducible representation V of G, as well as a Borel and a maximal torus. Let λ be the highest weight, let λ^* be the lowest weight, and let $\lambda = \lambda_1, \dots, \lambda_n = \lambda^*$ be all the weights.

Now pick a B-stable flag $0 = F_0 \subset \cdots \subset F_n = V$ so that each subquotient F_i/F_{i-1} is a 1-dimensional representation of T with weight λ_i. This gives a flag $\mathcal{F}_\bullet$ of locally free $\mathcal{O}_\mathcal{B}$-modules, where the fiber of $\mathcal{F}_i$ at $B \in \mathcal{B}$ is F_i. Each subquotient $\mathcal{F}_i/\mathcal{F}_{i-1}$ is $\mathcal{O}(\lambda_i)$. We also have $\mathcal{F}_n = V \otimes_\mathbb{C} \mathcal{O}_\mathcal{B}$.

Given an $\mathcal{O}_\mathcal{B}$-module $\mathcal{M}$ we may consider the corresponding twisted flag $\mathcal{F}_i(\mathcal{M}) = \mathcal{F}_i \otimes_{\mathcal{O}_\mathcal{B}} \mathcal{M}$. The subquotients are $\mathcal{M}(\lambda_i) = \mathcal{O}(\lambda_i) \otimes_{\mathcal{O}_\mathcal{B}} \mathcal{M}$. We also have $\mathcal{F}_n(\mathcal{M}) = V \otimes_\mathbb{C} \mathcal{M}$. There is a projection map $\mathrm{pr} : \mathcal{F}_n(\mathcal{M}) \to \mathcal{F}_n(\mathcal{M})/\mathcal{F}_{n-1}(\mathcal{M}) = \mathcal{M}(\lambda^*)$.

Lemma 7.1 (Splitting Lemma)**.** *If $\mathcal{M}$ is a $\mathcal{D}_\mathcal{B}$-module, then* pr *splits.*

Proof. We have a map $\mathcal{U}\mathfrak{g} \to \mathcal{D}(\mathcal{B})$, which allows $\mathcal{U}\mathfrak{g}$ to act on $\mathcal{M}$. Any z in the center $Z\mathfrak{g}$ of $\mathcal{U}\mathfrak{g}$ acts by $\chi_0(z)\mathrm{id} = \mathrm{hc}(z)(0)\mathrm{id}$, where hc is the Harish-Chandra isomorphism. There is also a twisted version of this, where for any character ν, we have $\mathcal{M}(\nu)$ is a $\mathcal{D}_{\mathcal{B},\mathcal{O}(\nu)}$-module, and $z \in Z\mathfrak{g}$ acts on it by $\chi_\nu(z)\mathrm{id} = \mathrm{hc}(z)(\nu)\mathrm{id}$.

To prove the splitting, it suffices to construct $z \in Z\mathfrak{g}$ such that $\chi_{\lambda_i}(z) \neq \chi_{\lambda_n}(z)$ for all $i \neq n$; this will produce a polynomial $p \in \mathbb{C}[t]$ such that $p(z)(\mathcal{F}_{n-1}(\mathcal{M})) = 0$ and $p(z)$ acts on the quotient $\mathcal{M}(\lambda^*)$ by id.

Recall that the HC isomorphism is a map $\mathrm{hc} : Z\mathfrak{g} \xrightarrow{\sim} (\mathcal{U}\mathfrak{t})^{\dot{W}} = \mathcal{O}(\mathfrak{t}^*/\dot{W})$, where χ_{λ_i} corresponds to evaluation at $\dot{W}\lambda_i$. ($\dot{W}$ is twisted dot action defined previously.) Elements of $\mathcal{O}(\mathfrak{t}^*/\dot{W})$ separate points of $\mathfrak{t}^*/\dot{W}$, so we are reduced to proving the following:

- If $i \neq n$, then $\lambda^* - \rho \notin W(\lambda_i - \rho)$.

Assume for contradiction that there is some $w \in W$ and $i \neq n$ such that $\lambda^* - \rho = w(\lambda_i - \rho)$. Then

$$(\rho - w(\rho)) + (w(\lambda_i) - \lambda^*) = 0 \tag{4}$$

Note that ρ and $-\lambda^*$ are dominant. Recall the partial order on $X^*(T)$: add. We have $w(\rho) \leq \rho$ with equality if and only if $w = 1$. We also have $\lambda^* \leq w(\lambda_i)$, since λ^* is the lowest weight. The equation 4 then forces $\rho = w(\rho)$ and $w(\lambda_i) = \lambda^*$, whence $w = 1$ and $\lambda_i = \lambda^*$, which contradicts λ^* being the lowest weight.

From 2 BB we get equivalences of categories

$$\mathcal{U}_0\text{-mod} \cong \mathcal{D}(\mathcal{B})\text{-mod} \cong \mathcal{D}_{\mathcal{B}}\text{-mod} \cong \mathcal{D}_\nu\text{-mod} \cong \mathcal{D}_\nu(\mathcal{B})\text{-mod} \cong \mathcal{U}_\nu(\mathcal{B})\text{-mod}$$

for any antidominant ν. Recall the notation $\mathcal{U}_0$ (resp. $\mathcal{U}_\nu$) means $\mathcal{U}\mathfrak{g}/(\chi_0)$ (resp. $\mathcal{U}\mathfrak{g}/(\chi_\nu)$).

Corollary 7.1. *For all antidominant ν, there is an equivalence (called* ***translation****)*

$$\mathcal{U}_0\text{-mod} \cong \mathcal{U}_\nu\text{-mod}.$$

In other words, $\mathcal{U}_\nu$ is Morita equivalent to $\mathcal{U}_0$.

The construction of something uses $\mathcal{D}(G/N)^G$. We consider $\mathcal{D}(G/N)^T$. We have

$$\mathcal{U}\mathfrak{g} \longleftarrow Z\mathfrak{g} \xrightarrow{\text{hc}} (\mathcal{U}\mathfrak{t})^{\dot{W}} \longrightarrow \mathcal{U}\mathfrak{t}$$

so we get an algebra map $\mathcal{U}\mathfrak{g} \otimes_{Z\mathfrak{g}} \mathcal{U}\mathfrak{t} \to \mathcal{D}(G/N)^T$. The LHS can also be described as the quotient of $\mathcal{U}\mathfrak{g} \otimes \mathcal{U}\mathfrak{t}$ by the relations $z \otimes 1 = 1 \otimes \text{hc}(z)$ for $z \in Z\mathfrak{g}$.

Claim 7.1. *The map $\mathcal{U}\mathfrak{g} \otimes_{Z\mathfrak{g}} \mathcal{U}\mathfrak{t} \to \mathcal{D}(G/N)^T$ is an isomorphism.*

Proof sketch. Taking associated graded with respect to the PBW filtration on both sides, we are reduced to proving that $\text{Sym}\mathfrak{g} \otimes_{(\text{Sym}\mathfrak{g})^G} \text{Sym}\mathfrak{t} \xrightarrow{\Psi} \mathcal{O}(T^*(G/N))^T$ is an isomorphism. The LHS can also be expressed as $\mathbb{C}[\mathfrak{g}^* \times_{\mathfrak{g}^*//G} \mathfrak{t}^*]$, so our morphism Ψ has a geometric incarnation as a map $T^*(G/N)//T \to \mathfrak{g}^* \times_{\mathfrak{g}^*//G} \mathfrak{t}^*$.

Recall that $T^*(G/N) = G \times^N \mathfrak{n}^\perp$. If we use a Killing form to identify $\mathfrak{g}^* \cong \mathfrak{g}$, then $\mathfrak{n}^\perp \cong \mathfrak{b}$. Now, T acts on this space by right translations, and the quotient is the Grothendieck-Springer resolution $\widetilde{\mathfrak{g}}$, which is the space of pairs $(x, \mathfrak{b}') \in \mathfrak{g} \times \mathcal{B}$ such that $x \in \mathfrak{b}'$. The map Ψ is induced by the map $\mu \times \nu : \widetilde{\mathfrak{g}} \to \mathfrak{g} \times_{\mathfrak{g}//G} \mathfrak{t}$, where $\mu(x, \mathfrak{b}') = x$ and $\nu(x, \mathfrak{b}') = x \mod \mathfrak{n}' \in \mathfrak{b}'/\mathfrak{n}' \cong \mathfrak{t}$. We have the following diagram

$$\begin{tikzcd} & \widetilde{\mathfrak{g}} \arrow[dl, "\mu"'] \arrow[dr, "\nu"] & \\ \mathfrak{g} \arrow[dr] & & \mathfrak{t} \arrow[dl] \\ & \mathfrak{g}//G \cong \mathfrak{t}//W & \end{tikzcd}$$

and we want to prove that $(\mu \times \nu)^*$ is an isomorphism. We have the following facts:

- $\mu \times \nu$ is birational and proper.

- $\mathfrak{g} \times_{\mathfrak{g}//G} \mathfrak{t}$ is a complete intersection in $\mathfrak{g} \times \mathfrak{t}$. Indeed, $\mathfrak{g}//G \cong \mathfrak{t}//W$ is an affine space with $\dim(\mathfrak{g}//G) = \dim(\mathfrak{t})$, so that many equations are needed to define the fiber product. Furthermore, $\dim(\mathfrak{g} \times_{\mathfrak{t}//W} \mathfrak{t}) = \dim(\mathfrak{g}) = \dim(\mathfrak{g} \times \mathfrak{t}) - \dim(\mathfrak{t})$, so the codimension of $\mathfrak{g} \times_{\mathfrak{g}//G} \mathfrak{t}$ in $\mathfrak{g} \times \mathfrak{t}$ equals the number of equations needed to define it.

Now we need $\mathfrak{g} \times_{\mathfrak{g}//G} \mathfrak{t}$ to be normal. It suffices to show that $\mathfrak{g} \times_{\mathfrak{g}//G} \mathfrak{t}$ is reduced and smooth in codimension 1 (should it be 2?). Since $\mathfrak{g} \times_{\mathfrak{g}//G} \mathfrak{t}$ is a complete intersection, to show it is reduced, it suffices to show it generically. We omit the remaining geometric steps.

7.2 BB theorem for projective space

Let $V = \mathbb{C}^n$, $\mathbb{P} = \mathbb{P}(V) = \mathbb{P}^{n-1}$. The group GL_n acts on V, and its subgroup SL_n acts on $\mathbb{P}$. Then elements of $\mathfrak{gl}_n$ give vector fields on V, and elements of $\mathfrak{sl}_n$ give vector fields on $\mathbb{P}$. Let x_i be coordinates on V, let $\partial_i = \frac{\partial}{\partial x_i}$, let $e_{ij} \in \mathfrak{g}$ be elementary matrices. The matrix e_{ij} is sent to the vector field $x_i \partial_j$ (or $x_j \partial_i$, depending on your convention) on V. One can show that the element

$$e_{ij}e_{k\ell} - e_{i\ell}e_{kj} - \delta_{jk}e_{i\ell} + \delta_{k\ell}e_{ij} \in \mathcal{U}\mathfrak{gl}_n$$

is mapped to 0 in $\mathcal{D}(V)$. (I verified it on paper.) Let $\mathcal{I}$ be the two-sided ideal generated by all elements of the above form, together with the identity matrix id.

Theorem 7.1. *The map $\mathcal{U}\mathfrak{gl}_n/\mathcal{I} \to \mathcal{D}(\mathbb{P})$ is an algebra isomorphism.*

Proof. Taking associated graded, we are reduced to showing that there is an isomorphism $\mathrm{Sym}\mathfrak{gl}_n/I \to \mathcal{O}(T^*\mathbb{P})$, where I is the ideal generated by the quadratic terms $e_{ij}e_{k\ell} - e_{i\ell}e_{kj}$, together with id. missing some stuff

8 Apr 16

8.1 Basic constructions with D-modules

Let $V = \mathbb{C}^n$ with coordinates x_i. Let $\partial_i = \frac{\partial}{\partial x_i}$. Any $v \in V$ gives a directional derivative operator ∂_v. Multiplication in $\mathcal{D}(V)$ induces an isomorphism $\mathbb{C}[V] \otimes \mathrm{Sym} V \to \mathcal{D}(V)$ of $\mathbb{C}[V]$-modules; explicitly the map sends a spanning element $f \otimes v_1 \cdots v_k$ to $m_f \partial_{v_1} \cdots \partial_{v_k}$. There is a similar isomorphism

$$\mathrm{Sym} V \otimes \mathbb{C}[V] \to \mathcal{D}(V) \tag{5}$$

given by swapping the order of $\mathrm{Sym}V$ and $\mathbb{C}[V]$. Thus, for any differential operator

$$u = \sum_{i_1+\cdots+i_n \leq k} m_{f_{i_1,\ldots,i_n}} \partial^{i_1} 1 \cdots \partial_n^{i_n},$$

there are unique functions $g_{i_1,\ldots,i_n}$ such that

$$u = \sum_{i_1+\cdots+i_n \leq k} \partial^{i_1} 1 \cdots \partial_n^{i_n} m_{g_{i_1,\ldots,i_n}}.$$

Now, let $I_0 = \mathbb{C}[V]_+$ be the augmentation ideal of those polynomials vanishing at 0. Let $\Delta_0 = \mathcal{D}(V)/\mathcal{D}(V)I_0$, called/thought of as "distributions at 0". The isomorphism 5 induces an isomorphism $\Delta_0 \cong \mathrm{Sym}V \otimes (\mathbb{C}[V]/I_0) = \mathrm{Sym}V$. Explicitly, $v_1 \cdots v_k \in \mathrm{Sym}V$ is sent to $\partial_{v_1} \cdots \partial_{v_k} \delta_0$, where δ_0 is 1 mod $\mathcal{D}(V)I_0$, thought of as the Dirac delta distribution.

Given a smooth algebraic variety X and a point $a \in X$, let $\mathcal{I}_a$ be the ideal sheaf of a. Then we define the $\mathcal{D}_X$-module $\Delta_a = \mathcal{D}_X/\mathcal{D}_X\mathcal{I}_a$. There is an induced filtration on Δ_a given by $F_i\Delta_a = \mathcal{D}_{X,i}\delta_a$, where δ_a is 1 mod $\mathcal{D}_X\mathcal{I}_a$.

Proposition 8.1. *1. Δ_a is a simple $\mathcal{D}_X$-module.*

2. There is a canonical isomorphism

$$\mathrm{gr}\Delta_a \cong \mathrm{Sym}(T_aX) = \mathbb{C}[T_a^*X].$$

3. $F_k\Delta_a$ consists of those $m \in \Delta_a$ with $I_a^{k+1}m = 0$.

4. $\mathrm{Supp}(\Delta_a) = \{a\}$*, and moreover, any $\mathcal{D}_X$-module supported on $\{a\}$ is a direct sum of copies of Δ_a. Note that* Supp *means support as an $\mathcal{O}_X$-module.*

Proof sketch. We only sketch the proof when $X = V = \mathbb{C}^n$ and $a = 0$, because later on we will prove something more general. There are two parts of the proof.

Part I: We have $F_k\Delta_0 \cong (\bigoplus_{i \leq k} \mathrm{Sym}^i V)\delta_0$. We have maps

$$\mathrm{Sym}^k V \underset{x_i}{\overset{\partial_i}{\rightleftarrows}} \mathrm{Sym}^{k+1} V$$

We compute $x_i\partial_i\delta_0 = (\partial_i x_i - 1)\delta_0 = -\delta_0$ and $\mathrm{Eu}(\delta_0) = -n\delta_0$, where $\mathrm{Eu} = \sum x_i\partial_i$ is the Euler vector field. Now there is an algebra isomorphism $\mathcal{D}(\mathbb{C}^n) \cong \mathcal{D}(\mathbb{C})^{\otimes n}$ and under this we have $\Delta_0(\mathbb{C}^n) \cong (\Delta_0(\mathbb{C}))^{\otimes n}$. We reduce to the case $n = 1$; we write x for x_1 and ∂ for ∂_1. Then $\Delta_0 = \mathbb{C}[\partial]\delta_0$. The previous diagram of maps becomes

$$\mathbb{C}\partial^k\delta_0 \underset{x}{\overset{\partial}{\rightleftarrows}} \mathbb{C}\partial^{k+1}\delta_0$$

Claim 8.1. $x\partial^{k+1}\delta_0 = c\partial^k\delta_0$ *for a nonzero constant* c*.*

Then for any nonzero $m \in \Delta_0$, there is some $k \geq 0$ and nonzero constant c such that $x^k m = c\delta_0$ and $x^{k+1}m = 0$. It follows that for any two nonzero $m, m' \in \Delta_0$, there is some polynomial p and $k \geq 0$ such that $m' = p(\delta)x^k m$, which shows that Δ_0 is simple. It is also clear that for any $m \in \Delta_0$, there is some k large enough so that $I_0^k m = 0$, which shows $\mathrm{Supp}(\Delta_0) = 0$.

Part II: For an arbitrary smooth X, we can first of course reduce to affines. Furthermore, using étale coordinates (explained below), we can reduce from the case of an affine to (essentially) the case of a vector space.

8.2 Étale coordinates

Let X be smooth affine. By Noether normalization, shrinking X if necessary, one can construct an étale map $F : X \to \mathbb{C}^n$; in particular (which is all we care about from the étale condition) $d_xF : T_xX \xrightarrow{\sim} T_{F(x)}\mathbb{C}^n$ for all $x \in X$. Let $\partial_i = (dF)^*(\frac{\partial}{\partial x_i})$. Then the ∂_i form a basis of TX with the property that $[\partial_i, \partial_j] = 0$ for all i, j. Thus any differential operator on X can be expressed in the form

$$\sum_{i_1+\cdots+i_n \leq k} f_{i_1,\ldots,i_n}(x)\partial_1^{i_1}\cdots\partial_n^{i_n}$$

for some $f_\bullet \in \mathcal{O}(X)$. Then one can work with differential operators in essentially the same way as with a vector space.

8.3 Twisted version

Let $\mathcal{L}$ be a line bundle, and recall the TDO sheaf $\mathcal{D}_{X,\mathcal{L}} = \mathcal{L} \otimes \mathcal{D}_X \otimes \mathcal{L}^{-1}$, with all tensor products over $\mathcal{O}_X$. Given $a \in X$ with ideal sheaf $\mathcal{I}_a$. We can define $\Delta_{a,\mathcal{L}} = \mathcal{D}_{X,\mathcal{L}}/\mathcal{D}_{X,\mathcal{L}}\mathcal{I}_a$.

Exercise: Let $X = \mathcal{B} = G/B$. Let $e = B$. For a character λ, we have a map $\mathcal{U}\mathfrak{g} \to \Gamma(\mathcal{B}, \mathcal{D}_{\mathcal{B},\mathcal{O}(\lambda)})$, so that $\Delta_{e,\mathcal{O}(\lambda)}$ can be viewed as a $\mathcal{U}\mathfrak{g}$-module. Show that it is a Verma module and find the highest weight.

8.4 Another construction

Proposition 8.2. *Let $\mathcal{M}$ be a $\mathcal{D}_X$-module. If $\mathcal{M}$ is coherent as an $\mathcal{O}_X$-module, then $\mathcal{M}$ is locally free as an $\mathcal{O}_X$-module.*

Proof. "To show something is locally free, it suffices to show that locally it is free" - Victor Ginzburg.

Fix $x \in X$ and let $\mathcal{O}_x$ be the local ring at x with maximal ideal $\mathfrak{m}_x$. Let $D = \mathcal{O}_x \otimes_{\mathcal{O}_X} \mathcal{D}_X$ and let $M = \mathcal{O}_x \otimes_{\mathcal{O}_X} \mathcal{M}$. Since $\mathcal{M}$ is coherent, M is finitely generated as an $\mathcal{O}_x$-module. Let $\overline{M} = M/\mathfrak{m}_x M$; it is a finite dimensional vector space over the residue field $\kappa(x) = \mathcal{O}_x/\mathfrak{m}_x$. Let $\overline{m_i}$ be a $\mathbb{C}$-basis for $\overline{M}$, and pick lifts $m_i \in M$. There is a map $p : \mathcal{O}_x^n \to M$ given by $(f_i) \mapsto \sum f_i m_i$. Nakayama's lemma implies p is surjective. The proof is done once we show the following:

Claim 8.2. *p is injective.*

Proof of Claim. Assume p is not injective. Define the order of a tuple (f_i) to be the maximum $k \geq 0$ such that each f_i is in $\mathfrak{m}_x^k$. Let ν be the minimum order of all tuples in $\ker p$, and take $(f_i) \in \ker p$ with order ν. Choose étale coordinates around $x \in X$ so that we have operators ∂_i.

Claim 8.3. *There is some $\partial_i = \partial$ such that $\partial f_1 \in \mathfrak{m}_x^{\nu-1}$ but not in $\mathfrak{m}_x^{\nu}$.*

We do not prove this claim. Assuming it, we have

$$0 = \partial \left(\sum f_i m_i \right) = \sum (\partial f_i) m_i + \sum f_i (\partial m_i).$$

By surjectivity of p, there are $a_{ij} \in \mathcal{O}_x$ such that $\partial m_i = \sum a_{ij} m_j$. Thus

$$0 = \sum (\partial f_i) m_i + \sum f_i a_{ij} m_j.$$

The coefficient of m_1 is $\partial f_1 + \sum f_i a_{i1}$, which is in $\mathfrak{m}_x^{\nu-1}$ but not in $\mathfrak{m}_x^{\nu}$. Thus we have produced a relation with order less than $\nu - 1$, a contradiction.

stuff at the end I don't quite understand

9 Apr 21

9.1 Review of connections

Let X be smooth, or holomorphic, or an algebraic variety. Let $\mathcal{O} = \mathcal{O}_X$ and Ω^i the sheaf of i-forms on X. For an $\mathcal{O}$-module $\mathcal{M}$, we can form $\mathcal{M} \otimes_{\mathcal{O}} \Omega^i$ to get $\mathcal{M}$-valued forms. We note $\mathcal{H}om_{\mathcal{O}}(T_X, \mathcal{M}) \cong \mathcal{M} \otimes \Omega^1$.

Definition 9.1. A **connection** on $\mathcal{M}$ is a map $\nabla : \mathcal{M} \to \mathcal{M} \otimes_{\mathcal{O}} \Omega^1$ such that $\nabla(fm) = m \otimes df + f\nabla m$.

Example 9.1. If $\mathcal{M} = \mathcal{O}$, we can take $\nabla = d$.

Identifying $\mathcal{H}om_{\mathcal{O}}(T_X, \mathcal{M}) \cong \mathcal{M} \otimes \Omega^1$, we can interpret ∇m as an element of $\mathcal{H}om_{\mathcal{O}}(T_X, \mathcal{M})$, denoted $\xi \mapsto \nabla_\xi m$.

Example 9.2. Let V be a vector space, $\mathcal{M} = \mathcal{O} \otimes_{\mathcal{C}} V$. Then $\nabla(f \otimes v) = (df) \otimes v + f\nabla(1 \otimes v)$, so the connection is determined by a map $A : X \to \mathrm{End} V$ (this isn't right but I'm not sure what was supposed to go here) (**connection 1-form**).

One can extend ∇:

$$\mathcal{M} \xrightarrow{\nabla} \Omega^1 \otimes \mathcal{M} \xrightarrow{\nabla} \Omega^2 \otimes \mathcal{M} \xrightarrow{\nabla} \cdots$$

However, in general $\nabla^2 \neq 0$.

Claim 9.1. $\nabla^2 \in (\mathrm{End}_{\mathcal{O}} \mathcal{M}) \otimes_{\mathcal{O}} \Omega^2$. *This is called the* ***curvature*** *of* ∇. *If* $\nabla^2 = 0$, *we say the connection is* ***flat***.

Example 9.3. Let $\mathcal{M} = \mathcal{O} \otimes_{\mathcal{C}} V$, $X = \mathbb{C}^n$ with coordinates x_i. The connection 1-form can be expressed as $A = \sum A_i dx_i$, with $A_i : X \to \mathrm{End} V$. Then

$$\nabla^2 = \sum_{i<j} \left(\frac{\partial}{\partial x_j} A_i - \frac{\partial}{\partial x_i} A_j + [A_i, A_j] \right) dx_i dx_j.$$

Remark. If $\dim V = 1$, then $\mathrm{End} V = \mathbb{C}$, so the connection 1-form A is a genuine 1-form, and $\nabla^2 = dA$.

Claim 9.2. *The map* $\xi \mapsto (m \mapsto \nabla_\xi m)$ *extends to a* $\mathcal{D}_X$*-module structure on* $\mathcal{M}$ *iff* $\nabla^2 = 0$.

Example 9.4. Let $\mathcal{M} = \mathcal{O}$. Fix $f \in \mathcal{O}$, and let $\nabla = d + A = d + df$. Then $\nabla_\xi(1) = \langle df, \xi \rangle \cdot 1 = \xi(f) \cdot 1$. Then the corresponding $\mathcal{D}$-module structure on $\mathcal{M}$ is exactly what we previously defined as $\mathcal{D}f$.

Example 9.5. Let $X = \mathbb{C}^\times$ with coordinate z, $\mathcal{M} = \mathcal{O}$. We have an invariant 1-form $z^{-1}dz$ and invariant vector field $\xi = z\frac{d}{dz}$. Fix $\lambda \in \mathbb{C}$ and let $\nabla = d + \lambda z^{-1} dz$. Then $\nabla_\xi(1) = \lambda \langle z^{-1}dz, \xi \rangle \cdot 1 = \lambda \cdot 1$. Note that $\xi(z^\lambda) = \lambda z^\lambda$. The resulting $\mathcal{D}$-module is denoted $\mathcal{D}z^\lambda$.

9.2 Pullback of D-modules

Given $F : X \to Y$ a morphism of varieties, we claim that there is a pullback functor $F^* : \text{-mod}\mathcal{D}_Y \to \mathcal{D}_X\text{-mod}$. At the level of $\mathcal{O}$-modules, it is exactly the $\mathcal{O}$-module pullback, i.e. $F^*\mathcal{M} = \mathcal{O}_X \otimes_{F^{-1}\mathcal{O}_Y} F^{-1}\mathcal{M}$, where F^{-1} denotes the naive sheaf pullback. However, it is not obvious why this has a $\mathcal{D}_X$-module structure, which we now explain in multiple ways.

The module $\mathcal{M}$ induces a flat connection $\nabla : \mathcal{M} \to \mathcal{M} \otimes_{\mathcal{O}_Y} \Omega^1_Y$. Pulling back the connection and composing with $F^*\Omega^1_Y \to \Omega^1_X$ gives a connection $F^*\nabla : F^*\mathcal{M} \to F^*\mathcal{M} \otimes_{\mathcal{O}_X} F^*\Omega^1_Y \to F^*\mathcal{M} \otimes_{\mathcal{O}_X} \Omega^1_X$. This connection is flat and thus gives a $\mathcal{D}_X$-module structure on $\mathcal{M}$.

We can also explain the D-module structure in terms of the action of vector fields. The map F induces $dF : T_X \to F^*T_Y$. Writing sections of $F^*\mathcal{M}$ as $f \otimes F^{-1}m$ (via the defining expression of $F^*\mathcal{M}$), we write the action of $\xi \in T_X$ as

$$\xi(f \otimes F^{-1}m) = \xi(f) \otimes F^{-1}m + f \otimes dF(\xi)m.$$

Example 9.6. Let $X = \mathbb{C}^k$ and $Y = \mathbb{C}^n$ with coordinates x_i and y_j respectively. Let $F : X \to Y$ be written in terms of coordinate functions $y_j(x)$. We can write

$$\frac{\partial}{\partial x_i} = \sum_j \frac{\partial y_j(x)}{\partial x_i} \frac{\partial}{\partial y_j}.$$

In particular,

$$\frac{\partial}{\partial x_i} : f(x) \otimes F^{-1}m \mapsto \frac{\partial f}{\partial x_i} \otimes F^{-1}m + \sum_j f(x) \frac{\partial y_j(x)}{\partial x_i} \frac{\partial}{\partial y_j} F^{-1}m.$$

Example 9.7 (Pullback along projection)**.** Let $F : X = Y \times Z \to Y$ be the projection. Then $F^*\mathcal{M} = \mathcal{M} \boxtimes \mathcal{O}_Z$. We note $\mathcal{D}_X = \mathcal{D}_Y \boxtimes \mathcal{D}_Z$, which naturally acts on $\mathcal{M} \boxtimes \mathcal{O}_Z$.

9.3 Pushforward of D-modules along an open embedding

Let $F : X \hookrightarrow Y$ be an open embedding. Then F_* for D-modules is the same as F_* for $\mathcal{O}$-modules, which is the same as F_* (naive pushforward) for sheaves of vector spaces. Let $\mathcal{M}$ be a sheaf on X, and let $U \subset Y$ be open. Then $F_*\mathcal{M}(U) = \mathcal{M}(U \cap X)$. The $\mathcal{D}_Y(U)$-module structure on $F_*\mathcal{M}(U)$ is then given by restricting to $\mathcal{D}_Y(U \cap X) = \mathcal{D}_X(U \cap X)$. We note that (in this case) F_* is left exact, but not exact in general.

9.4 Beilinson-Bernstein theorem for projective space

Let $\mathbb{P} = \mathbb{P}(\mathbb{C}^n) = \mathbb{P}^{n-1}$.

Theorem 9.1. *We have mutually inverse equivalences* $\Gamma = \mathrm{Loc}^{-1} : \mathcal{D}_{\mathbb{P}}\text{-mod} \to \mathcal{D}(\mathbb{P})\text{-mod}$.

Proof. It suffices to prove:

- Γ is exact.
- If $\mathcal{M}$ is a nonzero $\mathcal{D}_{\mathbb{P}}$-module, then $\Gamma(\mathcal{M}) \neq 0$.

First we prove statement 2.

Consider $p : \mathbb{C}^n \setminus 0 \to \mathbb{P}$; it is a principal $\mathbb{C}^\times$-bundle. Let $\mathcal{M}$ be a nonzero $\mathcal{D}_{\mathbb{P}}$-module. There is a decomposition $\Gamma(p^*\mathcal{M}) = \bigoplus_{i \in \mathbb{Z}} \Gamma(p^*\mathcal{M})^{(i)}$ into weight spaces for the $\mathbb{C}^\times$-action. Each weight space can be reinterpreted as $\Gamma(\mathbb{P}, \mathcal{M}(i))$. In particular, $\Gamma(\mathcal{M}) = \Gamma(p^{-1}\mathcal{M}) = \Gamma(p^*\mathcal{M})^{(0)}$. The following claim is then clearly stronger than what we want to prove:

Claim 9.3. *If $\mathcal{M}$ is a nonzero $\mathcal{D}_{\mathbb{P}}$-module, then $\Gamma(p^*\mathcal{M})^{(i)} \neq 0$ for all $i \geq 0$.*

Proof of Claim. We proceed by descending induction on i. For i very large, $\mathcal{O}(i)$ is very ample, so $\Gamma(p^*\mathcal{M})^{(i)} = \Gamma(\mathbb{P}, \mathcal{M}(i)) \neq 0$.

We now show that, assuming $\Gamma(p^*\mathcal{M})^{(i)} \neq 0$ and $i > 0$, we have $\Gamma(p^*\mathcal{M})^{(i-1)} \neq 0$. The Euler vector field $\mathrm{Eu} = \sum x_j \frac{\partial}{\partial x_j}$ on $\mathbb{C}^n \setminus 0$ is the (image of a) generator for the $\mathrm{Lie}(\mathbb{C}^\times)$-action. Furthermore, Eu acts on $\Gamma(p^*\mathcal{M})^{(i)}$ by $i \cdot \mathrm{id}$. Write ∂_k for $\frac{\partial}{\partial x_k}$. We compute $[\mathrm{Eu}, \partial_k] = -1 \cdot \partial_k$. This implies ∂_k maps the weight i space to the $i-1$ space. Let $\widetilde{m} \neq 0$ be in the weight i space. Suppose that $\partial_k \widetilde{m} = 0$ for all k. Then $\mathrm{Eu}\widetilde{m} = 0$, a contradiction. Thus there is some k such that $\partial_k \widetilde{m} \neq 0$, which gives a nonzero element in the weight $i-1$ space.

Now we prove statement 1, i.e. Γ is exact.

Let
$$0 \to \mathcal{M}_1 \to \mathcal{M}_2 \to \mathcal{M}_3 \to 0$$
be a SES of $\mathcal{D}_{\mathbb{P}}$-modules. Let $p : \mathbb{C}^n \setminus 0 \to \mathbb{P}$, and let $j : \mathbb{C}^n \setminus 0 \hookrightarrow \mathbb{C}^n$. Since p is smooth, the functor p^* is exact, so we get

$$0 \longrightarrow j_*p^*\mathcal{M}_1 \longrightarrow j_*p^*\mathcal{M}_2 \xrightarrow{\alpha} j_*p^*\mathcal{M}_3 \longrightarrow \mathrm{coker}\alpha \longrightarrow 0$$

Since $\mathbb{C}^n$ is affine, taking global sections on it is exact. Furthermore, taking weight spaces is exact, so we obtain:

$$0 \longrightarrow \Gamma(j_*p^*\mathcal{M}_1)^{(0)} \longrightarrow \Gamma(j_*p^*\mathcal{M}_2)^{(0)} \longrightarrow \Gamma(j_*p^*\mathcal{M}_3)^{(0)} \longrightarrow \Gamma(\mathrm{coker}\alpha)^{(0)} \longrightarrow 0$$

which is the same as

$$0 \longrightarrow \Gamma(\mathcal{M}_1) \longrightarrow \Gamma(\mathcal{M}_2) \longrightarrow \Gamma(\mathcal{M}_3) \longrightarrow \Gamma(\mathrm{coker}\alpha)^{(0)} \longrightarrow 0$$

Now, $\mathrm{coker}(\alpha)$ is supported at 0, so it is a direct sum of copies of Δ_0. We know Eu acts on Δ_0 with weights $-n, -n-1, -n-2, \ldots$, so there is no weight 0 space.

10 Apr 23

10.1 Filtered to graded formalism

Let R be a not necessarily commutative ring, which is (left) Noetherian. Let $t \in R$ be a central nonzerodivisor, so that we can define the localization $R_t = R[\frac{1}{t}]$ in the same way as for commutative rings. The natural map $R \to R_t$ is injective.

Definition 10.1. Let M be a finitely generated R_t-module. An R-submodule $L \subset M$ is called a **lattice** if

1. L is finitely generated as an R-module.
2. $\bigcup_{i \geq 0} \frac{1}{t^i} L = M$.

Example 10.1. Let $m_1, \ldots, m_r$ be generators of M over R_t. Then the R-submodule L generated by $m_1, \ldots, m_r$ is a lattice. In particular, any finitely generated R_t-module has a lattice.

Lemma 10.1. *Let M be finitely generated over R_t. Let $L \subset M$ be an R-submodule satisfying $\bigcup_i \frac{1}{t^i} L = M$, but not necessarily a lattice.*

1. *Let $L' \subset M$ be a lattice. Then there is some natural number n such that $L' \subset \frac{1}{t^n} L$. In particular, if L is a lattice, then there is some n such that $t^n L \subset L' \subset \frac{1}{t^n} L$.*
2. *Let $0 \to M' \to M \to M'' \to 0$ be a short exact sequence of R_t-modules, and let L be a lattice. Then $L' = M' \cap L$ is a lattice in M' and the image L'' of L in M'' is a lattice in M''. Furthermore, we have a short exact sequence $0 \to L' \to L \to L'' \to 0$ of R-modules.*

Definition 10.2. Let A be any ring. The **Grothendieck semigroup** $K^+(A)$ is the abelian semigroup generated by isomorphism classes $[N]$ of finitely generated A-modules N, with relation $[N] = [N'] + [N'']$ if there is a short exact sequence $0 \to N' \to N \to N'' \to 0$.

Lemma 10.2. *Let M be a finitely generated R_t-module with two lattices L, L'. Then $[L/tL] = [L'/tL']$ in $K^+(R/tR)$.*

Proof. First assume that L, L' are "adjacent" in the sense that $tL' \subset tL \subset L' \subset L$. Then

$$\begin{aligned}
[L/tL] &= [L/L'] + [L'/tL], \\
[L'/tL'] &= [L'/tL] + [tL/tL'].
\end{aligned}$$

Multiplication by t gives an isomorphism $L/L' \cong tL/tL'$, which completes the proof in this case.

In the general case, let $L_j = L + t^j L'$ for $j \in \mathbb{Z}$. Then L_j and L_{j+1} are adjacent lattices. For very large j, we have $L_j = L$, and for very small j, we have $L_j = t^j L'$. Finally, $L'/tL' \cong t^j L'/t^{j+1} L'$, so we can transfer from $[L/tL]$ to $[L'/tL']$ by using the case of adjacent lattices.

Putting the lemmas together, we have:

Proposition 10.1. *The assignment $M \mapsto L/tL$ for a lattice $L \subset M$ induces a well-defined semigroup homomorphism called **specialization**:*

$$\lim_{t\to 0} K^+(R_t\text{-mod}) \to K^+(R/tR\text{-mod}).$$

Proposition 10.2. *If $B = R/tR$ is a commutative algebra, then B has a canonical Poisson bracket $\{-,-\}$, i.e. a Lie bracket such that for any $b \in B$, the map $\{b,-\} : B \to B$ is a derivation.*

Proof. We construct the bracket, but we do not prove that it satisfies the relevant properties.

Let $b_1, b_2 \in B$. Let $r_1, r_2 \in R$ have $b_i = r_i + tR$. Then $r_1r_2 - r_2r_1 \in tR$. Since t is not a zerodivisor, there is a unique $r \in R$ such that $tr = r_1r_2 - r_2r_1$. We claim that $b = r + tR$ is independent of the choice of r_1, r_2. We define $\{b_1, b_2\} = b$.

Remark. The map $b \mapsto \{b,-\}$ is a Lie algebra homomorphism $(B, \{-,-\}) \to \mathrm{Der}B$.

Let k be a field, A a not necessarily commutative k-algebra.

Definition 10.3. A **(multiplicative) filtration** on A is a chain of subspaces

$$\cdots \subset A_{-1} \subset A_0 \subset A_1 \subset \cdots$$

such that

1. $1 \in A_0$.
2. $\bigcup_i A_i = A$.
3. $A_iA_j \subset A_{i+j}$.

An algebra with a filtration is a **filtered algebra**.

Definition 10.4. If A is filtered, then there are two graded algebras associated to it. There is the **associated graded algebra** $\mathrm{gr}A = \bigoplus_i A_i/A_{i-1}$, and there is the **Rees algebra** $RA = \sum_i t^iA_i \subset A[t,t^{-1}]$. It is a graded $k[t]$-algebra. The Rees algebra is isomorphic to $\bigoplus A_i$, where we forget the inclusions between the pieces of the filtration.

The Rees algebra satisfies the following properties:

- RA is flat over $k[t]$.
- $RA/tRA \cong \mathrm{gr}A$.
- $(RA)_t = A[t,t^{-1}]$.

Geometric motivation: If A is commutative, then so are $\mathrm{gr}A$ and RA, and we have

$$\begin{tikzcd} \mathrm{Spec}(A) \times (\mathbb{A}^1 \setminus 0) \arrow[r, hook] \arrow[d] & \mathrm{Spec}(RA) \arrow[d] & \mathrm{Spec}(\mathrm{gr}A) \arrow[l, hook'] \arrow[d] \\ \mathbb{A}^1 \setminus 0 \arrow[r, hook] & \mathbb{A}^1 & 0 \arrow[l, hook'] \end{tikzcd}$$

Example 10.2. Let X be a variety with closed subvariety Y, defined by an ideal sheaf $I = I_Y \subset \mathcal{O}_X$. Filter $\mathcal{O}_X$ by putting $\mathcal{O}_X$ in non-negative degrees, and I^i in degree $-i$ for $i > 0$. We have $\mathrm{gr}_{-i}\mathcal{O}_X = I^i/I^{i+1}$ for $i \geq 0$. If X, Y are both smooth, then $\bigoplus_{i\geq 0} I^i/I^{i+1} = \mathrm{Sym}(I/I^2) = \mathrm{Sym}(T_Y^* X) = k[T_Y X]$, where $T_Y X$ is the normal bundle. The previous diagram becomes

$$\begin{tikzcd} X \times (\mathbb{A}^1 \setminus 0) \arrow[r, hook] \arrow[d] & \mathrm{Spec}(R\mathcal{O}_X) \arrow[d] & T_Y X \arrow[l, hook'] \arrow[d] \\ \mathbb{A}^1 \setminus 0 \arrow[r, hook] & \mathbb{A}^1 & 0 \arrow[l, hook'] \end{tikzcd}$$

This shows that there is a deformation from X to the normal bundle.

We can extend the definition of K^+ to allow categories with exact sequences, e.g. $\mathrm{Coh}X$. We write $K^+(X)$ in place of $K^+(\mathrm{Coh}X)$.

Corollary 10.1. *We have semigroup homomorphisms* $K^+(X) \xrightarrow{\mathrm{pr}_1^*} K^+(X \times (\mathbb{A}^1 \setminus 0)) \xrightarrow{\lim_{t\to 0}} K^+(T_Y X)$.

From now on, a filtered algebra A has $A_{-1} = 0$. Let M be an A-module.

Definition 10.5. A **filtration** on M is a chain of subspaces

$$\cdots \subset M_{-1} \subset M_0 \subset M_1 \subset \cdots$$

such that

1. $\bigcup_i M_i = M$.
2. $A_i M_j \subset M_{i+j}$.
3. Each M_j is finitely generated as an A_0-module.
4. $M_j = 0$ for j small enough.

A module with a filtration is a **filtered module**.

Assume M is filtered. We can define the **associated graded module** $\mathrm{gr}M = \bigoplus M_i/M_{i-1}$, which is a graded $\mathrm{gr}A$-module, and the **Rees module** $RM = \sum_i t^i M_i$, which is a graded RA-module. We have

- $RM/tRM \cong \mathrm{gr}M$.

- $(RM)_t = M[t, t^{-1}]$.

Proposition 10.3. *The following conditions on a filtration of M are equivalent:*

1. $\mathrm{gr}M$ *is finitely generated over* $\mathrm{gr}A$.
2. RM *is finitely generated over* RA.
3. *There is an integer n such that $M_{i+n} = A_i M_n$ for $i \geq 0$.*

Definition 10.6. The filtration on M is called **good** if the equivalent conditions in the preceding proposition hold.

We can always give a good filtration on a finitely generated A-module by setting $M_i = A_i M$ for $i \geq 0$ and $M_i = 0$ for $i < 0$.

Proposition 10.4. *Assume* $\mathrm{gr}A$ *is Noetherian. Then*

1. *A and RA are Noetherian.*
2. *Let $0 \to M' \to M \to M'' \to 0$ be a short exact sequence of A-modules, and suppose M has a good filtration. Then the restrictions $M'_i = M' \cap M_i$ form a good filtration on M', and the images M''_i of M_i in M'' form a good filtration on M''. Moreover, we have a short exact sequence $0 \to \mathrm{gr}M' \to \mathrm{gr}M \to \mathrm{gr}M'' \to 0$.*

Theorem 10.1. *Assume* $\mathrm{gr}A$ *is Noetherian. Then the assignment $M \mapsto \mathrm{gr}M$, with respect to a good filtration on M, induces a semigroup homomorphism $K^+(A) \to K^+(\mathrm{gr}A)$.*

11 Apr 28

11.1 Reminder on commutative algebra

Let $B = \bigoplus_{i\geq 0} B_i$ be a graded and commutative ring, further assumed to be an integral domain which is finitely generated over a field k of characteristic 0 (in particular, $k \subset B_0$).

As a result, B_0 is finitely generated over k, and B is generated as a B_0-algebra by finitely many homogeneous elements from $B_{>0}$. Setting $X = \mathrm{Spec} B$ and $X_0 = \mathrm{Spec} B_0$, we have these are irreducible varieties over k. The maps $B_0 \hookrightarrow B$ and $B \twoheadrightarrow B/B_{>0} = B_0$ induce maps $p : X \to X_0$ and $i : X_0 \hookrightarrow X$. We identify X_0 with its image in X. We have $p \circ i = \mathrm{id}$.

The grading on B induces a $\mathbb{G}_m$-action on X, such that X_0 is the fixed point locus.

Let $M = \bigoplus_{i\in\mathbb{Z}} M_i$ be a finitely generated B-module. Since M is finitely generated, the grading on M is bounded below. Each M_i is a finitely generated B_0-module. The grading on M induces a $\mathbb{G}_m$-action on the associated coherent sheaf $\widetilde{M}$ on X. The support of M is a closed $\mathbb{G}_m$-stable subset of X. Let $d(m) = \dim \mathrm{Supp} M$.

For each irreducible component S of $\mathrm{Supp} M$ with $\dim S = d(M)$, we define the multiplicity $(M : S) \in \mathbb{Z}_{>0}$ as follows. S corresponds to a prime ideal $\mathfrak{p}$ of B. If $d(M) = \dim(X)$, then $B_\mathfrak{p}$ is a field, in particular the fraction field of B, and $M_\mathfrak{p}$ is a finite dimensional vector space over $B_\mathfrak{p}$. In this case we say $(M : S) = \dim_{B_\mathfrak{p}} M_\mathfrak{p}$. In general, we find that $M_\mathfrak{p}$ is killed by some power of $\mathfrak{p}B_\mathfrak{p}$, so it is finite length, and we let $(M : S)$ be the length.

11.2 Hilbert polynomial

Assume now that $B_0 = k$. Then each B_i and M_i is finite dimensional over k.

Fact: There is a unique polynomial $p(M, t) \in \mathbb{Q}[t]$ of the form

$$p(M, t) = c(M)\frac{t^{d(M)}}{d(M)!} + \text{ lower powers of } t$$

such that:

1. For n large enough,

$$p(M, n) = \sum_{i\leq n} \dim M_i.$$

2. For $\mathcal{C}$ be the set of irreducible components S of $\mathrm{Supp} M$ with $\dim S = d(M)$. Then

$$c(M) = \sum_{S \in \mathcal{C}} (M : S).$$

3. If $0 \to M' \to M \to M'' \to 0$ is a short exact sequence of graded B-modules, then $p(M, t) = p(M', t) + p(M'', t)$.

11.3 Filtered to graded

Let A be a not necessarily commutative algebra with filtration $0 = A_{-1} \subset A_0 \subset \cdots$ with $1 \in A_0$ such that $B = \mathrm{gr} A$ is a commutative finitely generated domain. Let M be an A-module of the form $M = AM_0$, where M_0 is a finitely generated A_0-module. Then the $M_i = A_i M_0$ give a good filtration on M. From last time, we have a homomorphism of semigroups $K^+(A) \to K^+(B)$.

Definition 11.1. The **associated variety**, or **singular support**, of M is $SS(M) = \mathrm{Supp}(\mathrm{gr} M) \subset \mathrm{Spec}(B)$.

Remark. It is important that we have a good filtration on M. In particular, $SS(M)$ is independent of the (good) filtration on M because of the map $K^+(A) \to K^+(B)$.

We let $d(m) = \dim SS(M)$, which is what we previously called $d(\mathrm{gr} M)$. For any irreducible component S of $SS(M)$ of dimension $d(M)$, we define $(M : S) = (\mathrm{gr} M : S)$.

Now let $0 \to M' \to M \to M'' \to 0$ be a short exact sequence of A-modules, such that M has a good filtration. We said last time that M' and M'' inherit good filtrations from M in a natural way, and we get a short exact sequence of the associated graded modules. In particular, we get $(M : S) = (M' : S) + (M'' : S)$, where on the right hand side, if $\dim(S) > d(M')$, we say $(M' : S) = 0$, and similarly for M''.

11.4 The case of D-modules

Let Y be smooth affine. Let $A = \mathcal{D}(Y)$ equipped with order filtration. Then $B = \mathrm{gr} A = k[T^*Y]$, so $X = \mathrm{Spec} B = T^*Y$. We note that $X_0 = \mathrm{Spec} B_0 = Y$ is the 0 section. Then any finitely generated $\mathcal{D}(Y)$-module M, we have $SS(M) \subset T^*Y$. $\mathbb{G}_m$ acts on T^*Y by dilation along the fibers of the projection $T^*Y \to Y$, and $SS(M)$ is $\mathbb{G}_m$-stable.

Example 11.1. Take $Y = \mathbb{C}^n$. Then $A_0 = k[Y]$, which we call $\mathcal{O}_Y$. We take $M = M_0 = \mathcal{O}_Y$, which is itself a good filtration. Then $\mathrm{gr} M = \mathcal{O}_Y$, thought of as a D-module via the construction $\mathcal{D}_Y(1)$. as well. Then $SS(M) = \mathrm{Supp}(\mathcal{O}_Y)$ is the zero section Y inside T^*Y.

Example 11.2. Pick $y \in Y$ and let $M = \Delta_y = \mathcal{D}/\mathcal{D}I_y$, where I_y is the ideal of vanishing of y. Equip M with the quotient filtration coming from the order filtration on $\mathcal{D}$. Then $\mathrm{gr}M = \mathrm{Sym}(T_yY) = k[T_y^*Y]$. In particular, $SS(M)$ is exactly T_y^*Y.

The preceding two examples exhibit essentially opposite phenomena of singular support.

Example 11.3. Let $Y = \mathbb{A}^1$ with coordinate t, and let $M = \mathcal{D}_Y(t^{-1})$. Then $SS(M)$ is the union of T_0^*Y and Y inside T^*Y, both with multiplicity 1.

Example 11.4. Let $Y = \mathbb{A}^1$ with coordinate t, and let $M = \mathcal{D}_Y(\log t)$. Then $SS(M)$ is set-theoretically the union of T_0^*Y and Y inside T^*Y, but with different multiplicities.

11.5 Another case

Consider the case of A which is generated by a finite dimensional subspace A_1 which contains k. Let $A_0 = k$, and for $i > 1$, let A_i be spanned by elements of the form $a_1 \cdots a_i$, where each $a_j \in A_1$. Then this is a filtration on A. Then $\mathrm{gr}A$ is commutative if and only if $[A_1, A_1] \subset A_1$, i.e. A_1 is a Lie subalgebra of A. We assume this is the case.

k is a central ideal in A_1. There is a surjection $F : \mathcal{U}(A_1)/(1_{\mathcal{U}} - 1_A) \to A$, and A_i is the image of ith term of the PBW filtration on $\mathcal{U}(A_1)$. The associated graded version of this map is a map $\mathrm{Sym}(A_1/k) \to \mathrm{gr}A$.

Now set $A = \mathcal{D}(\mathbb{C}^n)$, which we know is generated by coordinates x_i and partials ∂_i. In particular, if we take A_1 to be the k-span of $\{1, x_1, \ldots, x_n, \partial_1, \ldots, \partial_n\}$, then we are in the setting as before, and we obtain a filtration on A called the **Bernstein filtration**. The associated graded is $k[x_1, \ldots, x_n, \partial_1, \ldots, \partial_n]$, where all generators have degree 1. A_1 is the **Heisenberg Lie algebra** $\mathcal{H}$; it has basis $1, x_1, \ldots, x_n, \partial_1, \ldots, \partial_n$ such that 1 is central, the x_i commute with each other, the ∂_i commute with each other, and $[\partial_i, x_j] = \delta_{ij} \cdot 1$. The Lie algebra $\mathcal{H}$ fits into a short exact sequence $0 \to k \to \mathcal{H} \to k^{2n} \to 0$, where k and k^{2n} are abelian Lie algebras of the appropriate dimension. We note $\mathcal{H}$ is nilpotent, in particular $[[\mathcal{H}, \mathcal{H}], \mathcal{H}] = 0$.

12 Apr 30

We begin with the setup from the end of last time; namely, let A be an associative k-algebra with a finite dimensional subspace A_1 with the following properties:

- $k \subset A_1$.
- $[A_1, A_1] \subset A_1$.
- A_1 generates A as a k-algebra.

We define a filtration so that $A_0 = k$ and A_i is spanned by i-fold products of elements of A_i for $i > 1$. (here the field k is $\mathbb{C}$, but I don't feel like changing things to be consistent. I will only use $\mathbb{C}$ to distinguish from dummy variables named k.)

Let M be an A-module such that there is a finite dimensional subspace M_0 with $M = AM_0$. Then $M_i = A_i M_0$ is a good filtration on M.

Lemma 12.1. *Assume $[A_1, [A_1, ...]] = 0$ where we apply the bracket r times. If $a \in A_i$ is such that kills $aM_{ri} = 0$, then $aM = 0$.*

Proof. We define a "differential filtration" $Q_\bullet = Q_\bullet A$ inductively. We set

$$Q_0 = \{a \in A \mid [a, A_1] = 0\},$$

which is in fact the center $Z(A)$ since A_1 generates A. Then for $i \geq 0$, we set

$$Q_{i+1} = \{a \in A \mid [a, A_1] \in Q_i\}.$$

The hypothesis on A_1 implies that every element in A is in some Q_i. In particular, $A_k \subset Q_{kr}$. It remains to prove the following claim.

Claim 12.1. *If $a \in Q_k$ is such that $aM_k = 0$, then $aM = 0$.*

Proof. We induct on k. When $k = 0$, we have $a \in Z(A)$ such that $aM_0 = 0$. Then $aM = aAM_0 = AaM_0 = A0 = 0$. Now suppose $k > 0$ and $a \in Q_k$ has $aM_k = 0$. We have $M_k = A_1 M_{k-1}$, so $aa_1 M_{k-1} = 0$ for all $a_1 \in A_1$ and $aM_{k-1} \subset aM_k = 0$, so $aM_{k-1} = 0$. Thus $[a, a_1]M_{k-1} = 0$. By induction and definition of Q_k, this means $[a, a_1]M = 0$. In other words, the action of a on M commutes with the action of A_1 on M, and thus the action of A on M. Since $aM_0 \subset aM_k = 0$, we get $aM = aAM_0 = AaM_0 = 0$ as desired.

The following result should be thought of as a sort of uncertainty principle.

Proposition 12.1 (Generalized Bernstein inequality)**.** *Assume M (as defined above) is a faithful A-module. Then*

$$d(M) = \dim SS(M) \geq \frac{1}{2} d(A) := \frac{1}{2} \dim \operatorname{Spec}(\operatorname{gr} A).$$

Proof. We have the Hilbert polynomial

$$p(M,t) := p(\mathrm{gr}M,t) = c(M)\frac{t^{d(M)}}{d(M)!} + \text{ lower order terms}$$

and similarly $p(A,t)$. An element $a \in A_k$ sends M_{rk} to $M_{rk+k} = M_{(r+1)k}$. By the Lemma and the faithful hypothesis, we get an injection $A_k \to \mathrm{Hom}_{\mathbb{C}}(M_{rk}, M_{(r+1)k})$. In particular, we have

$$\dim A_k \leq \dim \mathrm{Hom}_{\mathbb{C}}(M_{rk}, M_{(r+1)k}) = \dim M_{rk} \dim M_{(r+1)k}.$$

For j very large, we have $p(M,j) = \dim M_j$, so for k very large, we have

$$p(A,k) \leq p(M,rk)p(M,(r+1)k).$$

As polynomials in k, the left hand side is degree $d(A)$, while the right hand side is degree $2d(M)$, so we have $d(A) \leq 2d(M)$ as desired.

12.1 For D-modules

Throughout this subsection, $A = \mathcal{D}(\mathbb{C}^n)$.

We saw last time that $A = \mathcal{D}(\mathbb{C}^n)$ fits into the above framework if we take A_1 to be the span of $1, x_i, \partial_j$ for all i, j. In particular $[A_1, [A_1, A_1]] = 0$, so we can take $r = 2$ to match the above notation. In fact we have $[A_i, A_j] \subset A_{i+j-2}$.

Lemma 12.2. *The algebra A is simple.*

Proof. We claim that for all $a \in A_i \setminus A_0$, where $i > 0$, we have $AaA = A$. By direct computation, if a contains x_j in its expression, then $[\partial_j, a] \in \mathcal{D}_{i-1}$ is nonzero. Similarly if a contains ∂_j, then $[x_j, a] \in \mathcal{D}_{i-1}$ is nonzero. Thus we can reduce $a \in AaA$ down to $1 \in AaA$, so we are done.

Proposition 12.2 (Bernstein inequality)**.** *For any nonzero finitely generated A-module M, we have $d(M) \geq n$.*

Proof. The annihilator of M is a two-sided ideal in A, so by the preceding Lemma we have that M is faithful. Since M is finitely generated, we can certainly write it in the form $M = AM_0$ for a finite dimensional subspace, namely taking M_0 to be the linear span of generators. Thus by the generalized Bernstein inequality, and the fact that $d(A) = 2n$, we are done.

Definition 12.1. We say an A-module is **holonomic** if $d(M) = n$, or if $M = 0$.

Proposition 12.3.

1. *If M is holonomic, then any submodule of M is holonomic, and any quotient of M is holonomic.*
2. *If $0 \to M' \to M \to M'' \to 0$ is a short exact sequence of holonomic modules, then $c(M) = c(M') + c(M'')$.*

12.2 b-function setup

Fix $f \in k[x_1, \ldots, x_n]$ of degree d. Let $U = k^n \setminus \{f = 0\}$. Then $\mathcal{O}(U) = k[x_1, \ldots, x_n, \frac{1}{f}]$. We will adjoin a formal central variable s to both $A = \mathcal{D}(k^n)$ and $\mathcal{D}(U)$. Let M_U be a $\mathcal{D}(U)$-module. We define $M_U f^s[s]$ a $\mathcal{D}(U)[s]$-module as follows. Elements have the form $\sum s^i m_i f^s$ for $m_i \in M_U$. Functions and powers of s act in the obvious way. A vector field ξ acts by

$$\xi \left(\sum s^i m_i f^s \right) = \sum s^i \left(\xi(m_i) + s\frac{\xi(f)}{f} m_i \right) f^s.$$

Now define an automorphism τ of $\mathcal{D}(U)[s]$ by identity on functions and $\xi \mapsto \xi + s\frac{\xi(f)}{f}$ on vector fields. Then for any $\mathcal{D}(U)[s]$-module M, we define a $\mathcal{D}(U)[s]$-module M^τ to be the underlying set M, but where an element a acts on an element m by $\tau(a)m$.

Claim 12.2. *$M_U f^s[s] \cong (M_U[s])^\tau$.*

13 May 5

13.1 b-function on affine space

We continue as before. Let $A = \mathcal{D}(\mathbb{C}^n)$ with Bernstein filtration. Let N be a $\mathcal{D}$-module with increasing filtration $F_\bullet N$ such that $A_i F_j N \subset F_{i+j} N$; we do not assume N is finitely generated or that the filtration is good.

Lemma 13.1. *Assume that for j large enough, $\dim(F_j N) \leq c j^d/d!+$ lower order terms where c, d are arbitrary. Then for any finitely generated submodule $M \subset N$, we have $d(M) \leq d$, and if $d = n$, then $d(M) = n$ and $c(M) \leq c_N$, where c_N is some constant depending on N.*

Proof. Since M is finitely generated, we can choose a good filtration M_i on M. Thus there is some large k such that $M_{i+k} = A_i M_k$ for all i. Let $F_j M = M \cap F_j N$. There is some large ℓ such that $M_k \subset F_\ell M$. Then for all i we have $M_{i+k} \subset F_{i+\ell} M$. Then for i large, we have

$$c(M)\frac{(i+k)^{d(M)}}{d(M)!} + \cdots = \dim M_{i+k} \leq \dim F_{i+\ell} M \leq \dim F_{i+\ell} N \leq c\frac{(i+\ell)^d}{d!} + \dots$$

so the result follows. might need some elaboration

Now, as last time, fix $f \in k[x_1, \dots, x_n]$ of degree d. Let $U = k^n \setminus \{f = 0\}$.

Theorem 13.1 (Theorem on b-function)**.** *Let $M = Am$ be a cyclic A-module which is f-torsion free, so $M \hookrightarrow M_f = M|_U$. If M is holonomic, then there is $b \in \mathbb{C}[s]$ and $u \in A[s]$ such that $u(mf^{s+1}) = b(s) m f^s$.*

Proof. Let $K = \mathbb{C}(s)$. Let $\widetilde{A} = K \otimes A = \mathcal{D}(K^n)$. Let $N = K \otimes M|_U f^s$, which is explicitly

$$\bigcup_{j \geq 0} \{\frac{1}{f^j} u m f^s \mid u \in \widetilde{A}\}.$$

We view N as a $\widetilde{A}$-module. We introduce the following filtration on N:

$$F_j N = \frac{1}{f^j}(\widetilde{A}_{j(d+1)} m) f^s.$$

The operator ∂_i acts on an element of $F_j N$ by

$$\partial_i : \frac{1}{f^j} u m f^s \mapsto \frac{1}{f^{j+1}} \left(f \partial_i u + (s-j)(\partial_i f) u\right) m f^s,$$

where $u \in \widetilde{A}_{j(d+1)}$. We want to show that the output is in $F_{(j+1)(d+1)} N$. Indeed, the term $f \partial_i u$ has degree at most $d+1+j(d+1) = (j+1)(d+1)$, and the term $(s-j)(\partial_i f) u$ has degree at most $d-1+j(d+1)$, which is less than $(j+1)(d+1)$. Now

$$\dim_K(F_j N) = \dim_K(\widetilde{A}_{j(d+1)} m) = \dim_{\mathbb{C}}(A_{j(d+1)} m).$$

For j large, we thus have

$$\begin{aligned}\dim_K(F_jN) &= c(M)\frac{(j(d+1))^{d(M)}}{d(M)!} + \cdots \\ &= c(M)(d+1)^n\frac{j^n}{n!} + \cdots\end{aligned}$$

where we use $d(M) = n$ by hypothesis that M is holonomic.

Now, for $j \geq 0$, we have a decreasing family of finitely generated $\widetilde{A}$-submodules $\widetilde{M}_j$ of N given by $\widetilde{M}_j = \widetilde{A}(mf^{s+j})$. By the Lemma, each $\widetilde{M}_j$ is a holonomic $\widetilde{A}$-module. Since holonomic modules have finite length, there is some large ℓ such that $\widetilde{M}_\ell = \widetilde{M}_{\ell+1}$. Thus there is some $\widetilde{u} \in \widetilde{A}$ such that $mf^{s+\ell} = \widetilde{u}(mf^{s+\ell+1})$. Via a translation of K, we may assume $\ell = 0$. Now we can write $\widetilde{u} = u/b(s)$ for some $u \in A[s]$, which gives the desired result.

Theorem 13.2. *Let M be a holonomic A-module. Then $M_f = M|_U$ is a holonomic A-module.*

Sketch. Since M has finite length, we may assume M is simple, and in particular, $M = Am$. Define $F_jM_f = \frac{1}{f^j}A_{j(d+1)}m$. Then a similar argument to the proof of the previous theorem suffices.

Note. "The meaning [of the filtration] is that Bernstein is very smart." - Victor Ginzburg

Example 13.1. Let $n = 1$ and $f(x) = x$ on $\mathbb{C}$. Then $U = \mathbb{C}^\times$. Let $M = \mathbb{C}[x] = A \cdot 1$, which is a holonomic A-module. The singular support of M is the zero section of $T^*\mathbb{C}$. We have $M_f = \mathbb{C}[x^{\pm 1}]$. This is in fact finitely generated as an A-module, namely $M_f = A \cdot \frac{1}{x}$. As an A-module, the quotient M_f/M is Δ_0. Since $SS(\Delta_0)$ is the fiber $T_0^*\mathbb{C}$, we see that $SS(M_f)$ is the "cross" inside $T^*\mathbb{C}$.

13.2 b-function on arbitrary variety

Let X be an arbitrary variety. Let $M = \mathcal{D}_X m$ be holonomic (now taken with respect to the order filtration) $\mathcal{D}_X$-module. Let $f \in \mathcal{O}(X)$ and $U = \{f \neq 0\}$.

Theorem 13.3. *There is some $u \in \mathcal{D}_X[s]$ and $b \in \mathbb{C}[s]$ such that $b(s)mf^s = u(mf^{s+1})$.*

Sketch. As before, let $K = \mathbb{C}(s)$ and form appropriate base changes. In particular, since M is holonomic, $\widetilde{\mathcal{D}}_U mf^s$ is what goes here?. Now we blackbox the following result:

Theorem 13.4 (Extension theorem)**.** *Let L be a finitely generated $\mathcal{D}_U$-module such that $\dim SS(L) = d$. Then there is a finitely generated $\mathcal{D}_X$-module E such that $E|_U = L$ and $\dim SS(E) = \dim SS(L)$.*

We extend $\widetilde{\mathcal{D}}_U mf^s$ to a $\widetilde{\mathcal{D}}_X$-module E. some quotient is supported on $\{f = 0\}$, so by Nullstellensatz, there is some large N such that f^N that kills it. In particular, $mf^{s+N} \in E$. Define $M_\ell = \mathcal{D}_X mf^{s+\ell}$ and then what?

13.3 Gabber-Kashiwara-Sato filtration

Let A be a filtered algebra such that $A_0 = k$, $\mathrm{gr}A$ is commutative, and $X = \mathrm{Spec}(\mathrm{gr}A)$ is Cohen-Macaulay. Let M bea finitely generated A-module with $d = d(M) = \dim SS(M)$.

Theorem 13.5. *There is a canonical filtration of M by A-submodules $G_i^{GKS}M$, for $i \in \{0, \dots, d\}$, such that*

- $\dim SS(G_i^{GKS}M) = i$ *for each i.*
- *All irreducible components of $SS(G_i^{GKS}M/G_{i-1}^{GKS}M)$ have dimension i.*

Let $\mathfrak{g}$ be a semisimple Lie algebra. Let $Z\mathfrak{g}$ be the center of $\mathcal{U}\mathfrak{g}$, and let $\mathcal{U}_0\mathfrak{g}$ be the quotient of $\mathcal{U}\mathfrak{g}$ by the augmentation ideal of $Z\mathfrak{g}$. Beilinson-Bernstein theorem says $\mathcal{U}_0\mathfrak{g} \cong \mathcal{D}(\mathcal{B})$. We have $\mathrm{Spec}(\mathrm{gr}\mathcal{U}_0\mathfrak{g}) = \mathcal{N} \subset \mathfrak{g}^*$. For a finitely generated $\mathcal{U}_0\mathfrak{g}$-module M, we may view M as a $\mathcal{D}(\mathcal{B})$-module, and then apply localization to get a $\mathcal{D}_{\mathcal{B}}$-module $\mathrm{Loc}M$. The singular support of $\mathrm{Loc}M$ sits inside $T^*\mathcal{B}$ and maps down to $SS(M) \subset \mathcal{N}$ under the Springer resolution.

14 May 7

14.1 Gabber's theorem

Let A be a non-negatively filtered algebra probably with $A_0 = k$ such that $\mathrm{gr}A$ is commutative and $X = \mathrm{Spec}(\mathrm{gr}A)$ is smooth and irreducible of dimension d. The assumption that X is smooth is critical for Gabber's theorem discussed below.

Recall that there is a canonical Poisson bracket on $\mathrm{gr}A$ defined by taking $(a, b) \in A_i \times A_j$ to $[\widetilde{a}, \widetilde{b}] \mod A_{i+j-2}$, where $\widetilde{a} \in A_i$ and $\widetilde{b} \in A_j$ are any lifts of a, b to A. In particular, X is a Poisson variety.

Definition 14.1. A closed subvariety $Z \subset X$ is called **coisotropic** if its defining ideal I_Z is closed under the Poisson bracket.

Remark. If X is symplectic, then Z is coisotropic if and only if for all smooth points $z \in Z$, the tangent space $T_z Z$ contains its orthogonal space with respect to the symplectic form. In particular, every irreducible component of Z has dimension at least $d/2$.

Example 14.1. Say $A = \mathcal{D}(Y)$ (with order filtration) so $X = T^*Y$ is symplectic, in which case the previous remark applies.

Theorem 14.1 (Gabber)**.** *For any finitely generated A-module M, the subvariety $SS(M)$ is coisotropic.*

We do not prove it; the proof is hard.

Corollary 14.1. *By taking $A = \mathcal{D}(Y)$, we recover Bernstein's inequality.*

Example 14.2. Let $I \subset A$ be a left ideal. Let $I_j = A_j \cap I$. Then $\mathrm{gr}I$ is an ideal in $\mathrm{gr}A$, which gives a subvariety $Z \subset X$. In fact, $SS(A/I) = Z$. Gabber's theorem says that Z is coisotropic. It is easy to check that $\mathrm{gr}I$ is stable under the Poisson bracket. However, Gabber's theorem says in fact that the radical of $\mathrm{gr}I$ is stable under the bracket, which is nontrivial.

Example 14.3. Let I be (left) generated by elements $a, b \in A$. Then $\mathrm{gr}I$ contains the ideal generated by the images of a, b in $\mathrm{gr}A$, but it can be larger. For instance, if we take $A = \mathcal{D}(\mathbb{C})$ and our elements to be x, ∂, then $I = A$ because $x\partial - \partial x = \pm 1 \in I$, but the images of x and ∂ generate a proper ideal in $\mathrm{gr}A$.

14.2 Symplectic geometry arising from our filtered algebra

Recall that commutativity of $\mathrm{gr}A$ is equivalent to A_1 being a Lie algebra with respect to the natural commutator.

Now let $\mathfrak{h}$ be a finite dimensional Lie subalgebra of A_1. Then there is a map $\mathcal{U}\mathfrak{g} \to A$ compatible with filtrations (PBW on the left). This it induces a map $\mathrm{gr}(\mathcal{U}\mathfrak{h}) \to \mathrm{gr}A$, i.e. $\mathbb{C}[\mathfrak{h}^*] \to \mathbb{C}[X]$. Thus we get a map $\mu : X \to \mathfrak{h}^*$, called the **moment map**.

Since A_1 is stable under commutator, we have $[A_1, A_i] \subset A_i$. Thus for all $a \in A_1$, the map $\{a \mod A_0, -\}$ is a derivation on $\mathbb{C}[X]$, thus determining a vector field on X. Such a vector field is called **Hamiltonian**. In particular, if we do this for our $\mathfrak{h}$, we get a map $\mathfrak{h} \to T_X$.

Let $(A, \mathfrak{h})$-mod be the category of finitely generated A-modules M such that the action of $\mathfrak{h}$ on M is locally finite, i.e. for all $m \in M$, there is a finite dimensional $\mathfrak{h}$-stable subspace $M_0 \subset M$ with $m \in M_0$.

Lemma 14.1. *For all $M \in (A, \mathfrak{h})$-mod, we have $SS(M) \subset \mu^{-1}(0)$.*

Proof. Choose generators $m_1, \ldots, m_r$ for M. Let M_0' be the $\mathbb{C}$-span of $m_1, \ldots, m_r$, and let $M_i' = A_i M_0'$. For each i, let E_i be a finite dimensional $\mathfrak{h}$-stable subspace containing m_i. Let $E = E_1 + \cdots + E_r \supset M_0'$, which is still finite dimensional and $\mathfrak{h}$-stable. Let $M_i = A_i E$; this is a good filtration of M such that $\mathfrak{h} M_i \subset M_i$. Since $\mathfrak{h} \subset A_1$, this implies $\mathfrak{h}$ acts by 0 on $\mathrm{gr}M$. Now, $\mu^{-1}(0)$ is (as a set) the variety defined by the ideal generated by the image of $\mathfrak{h}$ under $\mathfrak{h} \subset \mathbb{C}[\mathfrak{h}^*] \to \mathbb{C}[X]$. Thus we see $SS(M) = \mathrm{Supp}(\mathrm{gr}M)$ is contained in $\mu^{-1}(0)$.

14.3 A special case

Let $\mathfrak{g}$ be a finite dimensional Lie algebra with a Lie subalgebra $\mathfrak{h}$. Let $A = \mathcal{U}\mathfrak{g}$, so $X = \mathfrak{g}^*$. The moment map $\mu : \mathfrak{g}^* \to \mathfrak{h}^*$ is just restriction to $\mathfrak{h}$. Now let $\lambda \in \mathfrak{h}^*$ be a character, i.e. trivial on $[\mathfrak{h}, \mathfrak{h}]$. Let $M = \mathrm{Ind}_{\mathcal{U}\mathfrak{h}}^{A} \lambda = A \otimes_{\mathcal{U}\mathfrak{h}} \lambda$, i.e. the quotient of A by the left ideal generated by elements of the form $h - \lambda(h)$ for $h \in \mathfrak{h}$. Then $M \in (A, \mathfrak{h})$-mod. Thus $SS(M) \subset \mu^{-1}(0) = \mathfrak{h}^\perp \subset \mathfrak{g}^*$. In fact, $\mathrm{gr}M = \mathrm{Sym}(\mathfrak{g}/\mathfrak{h}) = \mathbb{C}[(\mathfrak{g}/\mathfrak{h})^*] = \mathbb{C}[\mathfrak{h}^\perp]$.

Note. When $\mathfrak{h}$ is a Borel, M is a Verma module.

Now suppose $\mathfrak{g}$ is semisimple, coming from some G. Let $\mathcal{U}_0$ be the quotient of $\mathcal{U}\mathfrak{g}$ by the augmentation ideal of $Z\mathfrak{g}$. Then $\mathrm{gr}\mathcal{U}_0 = \mathbb{C}[\mathcal{N}]$ where $\mathcal{N} \subset \mathfrak{g}^*$ is the nilpotent cone. It is a standard fact that $\mathcal{N}$ has finitely many orbits under the (co)adjoint action of G; these are called nilpotent orbits.

Theorem 14.2. *Let M be a simple $\mathcal{U}_0$-module.*

1. *The variety defined by* $\mathrm{grAnn}M$*, which is equivalently $SS(\mathcal{U}_0/\mathrm{Ann}M)$, is the closure of a single nilpotent orbit $\mathcal{O}$.*
2. $\dim SS(M) = \frac{1}{2}\dim\mathcal{O}$*. In fact this is true of all irreducible components of $SS(M)$.*

Let us comment on this result. Let I be a two-sided ideal of $\mathcal{U}\mathfrak{g}$. Then both I and $\mathrm{gr}I$ are stable under the adjoint action of $\mathfrak{g}$ on $\mathcal{U}\mathfrak{g}$. Geometrically, this means that the vector fields given by the adjoint action of $\mathfrak{g}$ on $\mathfrak{g}^*$ are all tangent to the variety defined by gr$I$. Supposing that we could exponentiate, this implies that the variety defined by gr$I$ is G-stable, and thus a union of orbits.

We continue assuming $\mathfrak{g}$ is semisimple, coming from some group G. Let $B \subset G$ and $\mathfrak{b} \subset \mathfrak{g}$ be Borels. We consider the category $(\mathcal{U}_0, \mathfrak{b})$-mod.

Note. This category is similar (equivalent as categories?) to category $\mathcal{O}$ with central character zero.

For any $M \in (\mathcal{U}_0, \mathfrak{b})$-mod, we know $SS(M) \subset \mu^{-1}(0)$ where $\mu : \mathcal{N} \to \mathfrak{b}^*$ is the restriction of the restriction map $\mathfrak{g}^* \to \mathfrak{b}^*$. In particular $\mu^{-1}(0) = \mathcal{N} \cap \mathfrak{b}^{\perp}$. If we identify $\mathfrak{g}$ with $\mathfrak{g}^*$ via a Killing form, then $\mathfrak{b}^{\perp}$ is identified with $\mathfrak{n} = [\mathfrak{b}, \mathfrak{b}]$, which sits inside $\mathcal{N}$, so $SS(M) \subset \mathfrak{n}$.

Now, consider the Springer resolution $\pi : T^*\mathcal{B} \to \mathcal{N}$. We have that $SS(\mathrm{Loc}M)$ sits inside $T^*\mathcal{B}$, and $SS(M)$ sits inside $\mathfrak{n} \subset \mathcal{N}$. It is a fact that $\pi^{-1}(\mathfrak{n})$ is the disjoint union of conormal bundles $T^*_{\mathcal{B}_w}\mathcal{B}$ to the Bruhat cells $\mathcal{B}_w$ for $w \in W$. The irreducible componenets of $\pi^{-1}(\mathfrak{n})$ are the closures of these conormal bundles, which are Lagrangian. Since π must map $SS(\mathrm{Loc}M)$ to $SS(M)$, we obtain:

Corollary 14.2. *For all $M \in (\mathcal{U}_0, \mathfrak{b})$-mod, the module $\mathrm{Loc}M$ is holonomic.*

15 May 12

15.1 Perfect obstruction theory (or -1 shifted (derived) symplectic geometry)

Ideas from paper by Brav, Bussi, Joyce https://arxiv.org/abs/1305.6302v4. Also related to a recent seminar talk at UChicago about microlocal stuff.

Let X be a smooth variety with closed and possibly singular subvariety Z. We let I be the ideal sheaf of Z.

The **normal bundle** is $N_Z = \mathrm{SpecSym}(I/I^2)$, which is a closed conical subvariety of $T_X|_Z$, where conical means stable under the fiberwise $\mathbb{C}^\times$-action. For singular I, this space may be poorly behaved. A better option is the **normal cone** $C_Z \subset N_Z$ given by $\mathrm{Spec}(\bigoplus_{i\geq 0} I^i/I^{i+1})$, where $I^0 = \mathcal{O}_X$. There is a natural map $\mathrm{Sym}(I/I^2) \to \bigoplus_{i\geq 0} I^i/I^{i+1}$, which is surjective by graded Nakayama's lemma, that induces the inclusion $C_Z \hookrightarrow N_Z$. If Z is smooth, the map $\mathrm{Sym}(I/I^2) \to \bigoplus_{i\geq 0} I^i/I^{i+1}$ is an isomorphism, so $C_Z = N_Z$. I don't remember if this is if and only if

Example 15.1. Let $Z = \{x_1^2 + x_2^2 + x_3^2 = 0\} \subset \mathbb{C}^3$. Then Z is singular only at the origin 0, and $N_{Z,0} = \mathbb{C}^3$. However, $C_{Z,0} = Z$ inside $N_{Z,0}$.

Let α be a 1-form on X, which is a section $X \to T^*X$. We take Z to be the zero locus of α, which is $\mathrm{im}(\alpha) \cap T^*_X X$. More explicitly, we have contraction $i_\alpha : T_X \to \mathcal{O}_X$ given by $\xi \mapsto \alpha(\xi)$, which is $\mathcal{O}_X$-linear, and we let Z be the subscheme determined by the ideal sheaf $\mathrm{im}(i_\alpha)$.

Now consider the composition $T_X \xrightarrow{i_\alpha} \mathcal{O}_X \xrightarrow{d} T_X^*$.

Theorem 15.1. *Assume $d\alpha = 0$. Then*

1. *$d \circ i_\alpha$ descends to an $\mathcal{O}_Z$-linear map $\mathrm{ob}_\alpha : T_X|_Z \to T_X^*|_Z$. This is called the **obstruction map**.*
2. *ob_α is self-adjoint.*
3. *The image of the composite*

$$C_Z \hookrightarrow N_Z \hookrightarrow T_X|_Z \xrightarrow{\mathrm{ob}_\alpha} T_X^*|_Z \hookrightarrow T^*X$$

is a conical Lagrangian subvariety.

Special case/local consideration: $\alpha = df$ for some $f \in \mathcal{O}(X)$. Then Z is the critical locus of f, i.e. set-theoretically those $x \in X$ with $d_x f = 0$. On this locus, there is a symmetric bilinear form $d^2 f$ on $T_X|_Z$. If x_i are étale coords, then $\frac{\partial^2 f}{\partial x_i \partial x_j}$ give the matrix entries. The obstruction map in this case is the Hessian $\frac{\partial}{\partial x_i} \mapsto \sum \frac{\partial^2 f}{\partial x_i \partial x_j} dx_j$.

Now let ω be the standard symplectic form on T^*X, and let Eu be the Euler vector field on T^*X that generates the fiberwise $\mathbb{C}^\times$-action. If we pick coordinates x_i for X and dual coordinates y_i for the fibers of $p : T^*X \to X$, then $\omega = \sum dx_i dy_i$ and $\mathrm{Eu} = \sum y_i \frac{\partial}{\partial y_i}$. The Liouville 1-form on T^*X is $\lambda = i_{\mathrm{Eu}}\omega$, which in coordinates is $\sum y_i dx_i$. Note that $\omega = -d\lambda$. This also follows, without coordinates, by Cartan's "magic formula".

Lemma 15.1. *Let $\Lambda \subset T^*X$ be a conical closed irreducible subvariety, so that $p(\Lambda) \subset X$. Then*

1. *The regular locus Λ_{reg} is Lagrangian if and only if $\dim \Lambda = \dim X$ and $\lambda = 0$ on Λ_{reg}.*
2. *If Λ_{reg} is Lagrangian, then $\Lambda = \overline{T^*_{p(\Lambda)_{\mathrm{reg}}} X}$.*

Now fix $z \in Z$, a critical point of f.

Claim 15.1. *The following sequence of vector spaces is exact:*

$$0 \to T_z Z \to T_z X \xrightarrow{\mathrm{ob}_{df}} T_z^* X \to T_z^* Z \to 0.$$

In other words, the induced map $\mathrm{ob}_{df} : N_z \to N_z^*$ is an isomorphism.

Theorem 15.2. 1. *$f(Z)$ is a finite set.*

2. *Let F be an automorphism of X such that $f \circ F = f$ and $F|_Z = \mathrm{id}$. Then the linear map $d_z F : T_z X \to T_z X$ has determinant ± 1.*

Now we discuss motivation from mirror symmetry. We want to compute GW = Gromov-Witten invariant(s). This involves some "virtual" classes that try to force things to behave as if a relevant space were 0-dimensional.

Key point: our Z, the critical locus of f, has virtual dimension zero.

Theorem 15.3. *There is a canonical constructible (i.e. constant on strata for some stratification) function $\nu_{Z,f} : Z \to \mathbb{Z}$.*

The relevant integers in GW are of the form

$$\sum_{k \in \mathbb{Z}} k \chi(\nu_{Z,f}^{-1}(k)),$$

where χ is the Euler characteristic.

Riemann-Hilbert is a functor dR from the category of holonomic $\mathcal{D}_X$-modules to the constructible derived category of X. Given a constructible complex $\mathcal{F}$, there is an associated constructible function $\eta(\mathcal{F})$ given by

$$\eta(\mathcal{F})(x) = \sum_{i \in \mathbb{Z}} (-1)^i \dim \mathcal{H}^i_x \mathcal{F}.$$

Now let $\mathcal{M}$ be a holonomic $\mathcal{D}_X$-module. To it, we can associate the constructible function $\eta(dR(\mathcal{M}))$, as well as its characteristic cycle. The characteristic cycle is the support cycle of $\mathrm{gr}\mathcal{M}$ with respect to a good filtration, and which is a conical Lagrangian cycle in T^*X. It turns out there is a construction which takes conical Lagrangian cycles in T^*X to constructible functions on X, which does not involve D-modules at all, but it is very complicated.

Theorem 15.4. *The image of C_Z in T^*X determines a conical Lagrangian cycle whose associated constructible function is $\nu_{Z,f}$.*

Remark. Furthermore, there is a $\mathcal{D}_X$-module that produces the cycle and constructible function, namely the $\mathcal{D}_X$-module of vanishing(?) cycles, which we will explain in due time.

Recall that Z is the intersection of two smooth Lagrangian subvarieties of T^*X, namely $\mathrm{im}(df)$ and the zero section $X \subset T^*X$.

Let M be a smooth symplectic variety with smooth Lagrangian subvarieties Λ_1, Λ_2. Set $Z = \Lambda_1 \cap \Lambda_2$. One can define obstruction and constructible function depending on Λ_1, Λ_2.

All of this has a 2-categorical perspective, at least according to physicists. There is a 2-category whose objects are Lagrangian subvarieties, 1-morphisms are called "matrix factorizations", and 2-morphisms are vector spaces that are essentially cohomology of vanishing cycles.

For E_1, E_2 vector bundles on X, there is a related moduli space of maps $u : E_1 \to E_2$, $v : E_2 \to E_1$, such that $uv = vu = f \cdot \mathrm{id}$.